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(54) **STATUS INDICATIONS FOR UPLINK CONFIGURED GRANT INSTANCES**

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H04W 72/1268 (2023.01)

(52) **U.S. Cl.**
CPC ***H04W 72/23*** (2023.01); ***H04W 72/1268*** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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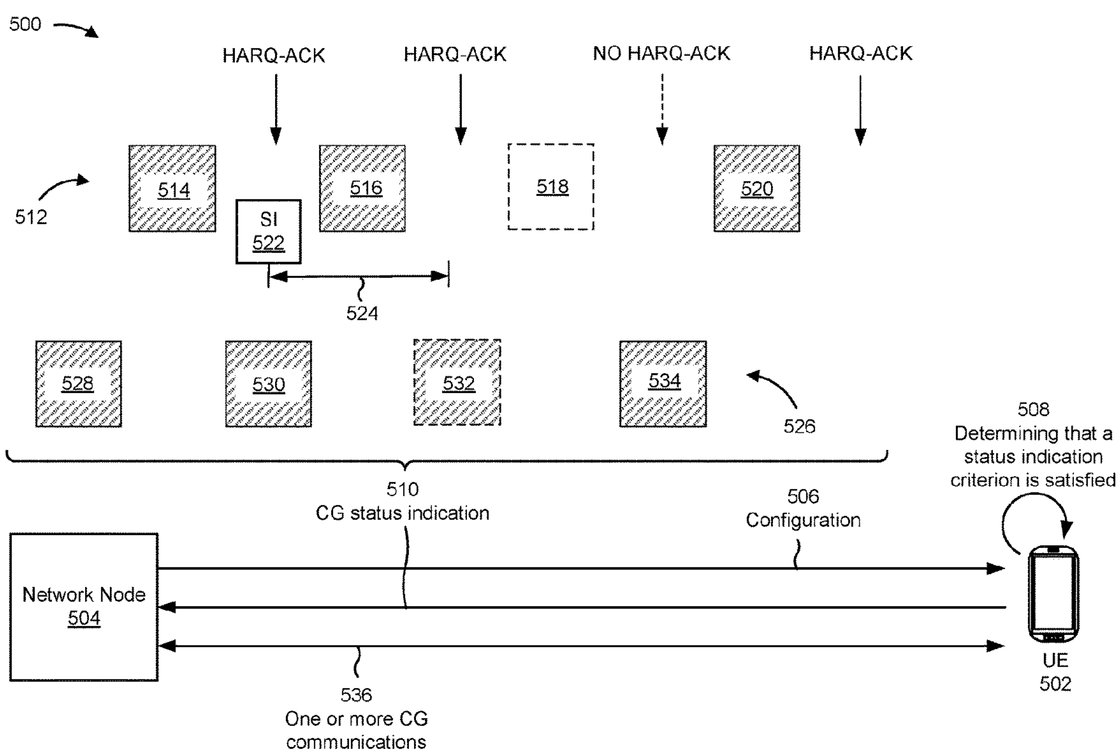
(57) **ABSTRACT**

Various aspects of the present disclosure generally relate to wireless communication. In some aspects, a user equipment (UE) may transmit a configured grant (CG) status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance. The UE may transmit one or more CG communications based at least in part on the status of the at least one CG instance. Numerous other aspects are described.

30 Claims, 9 Drawing Sheets



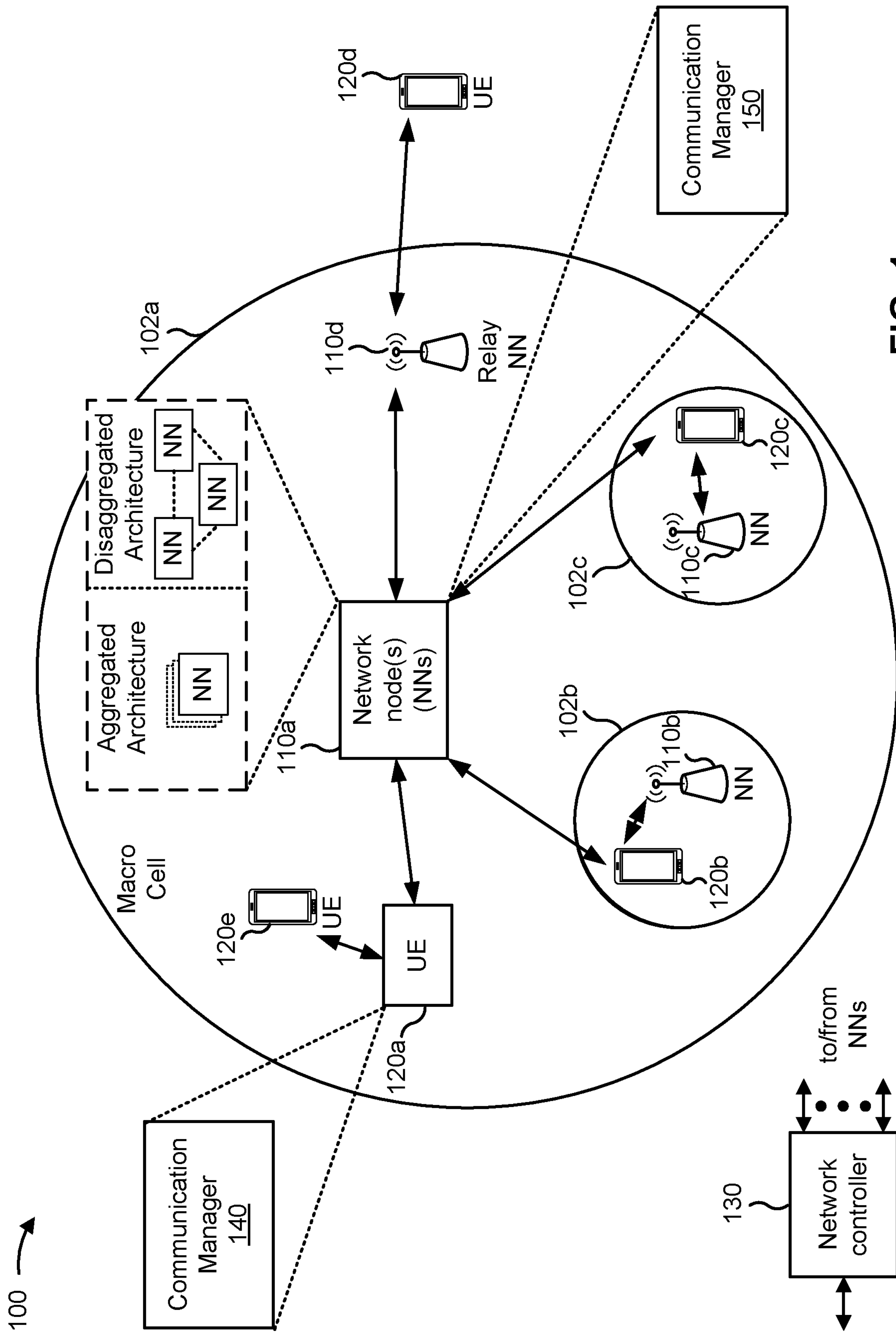
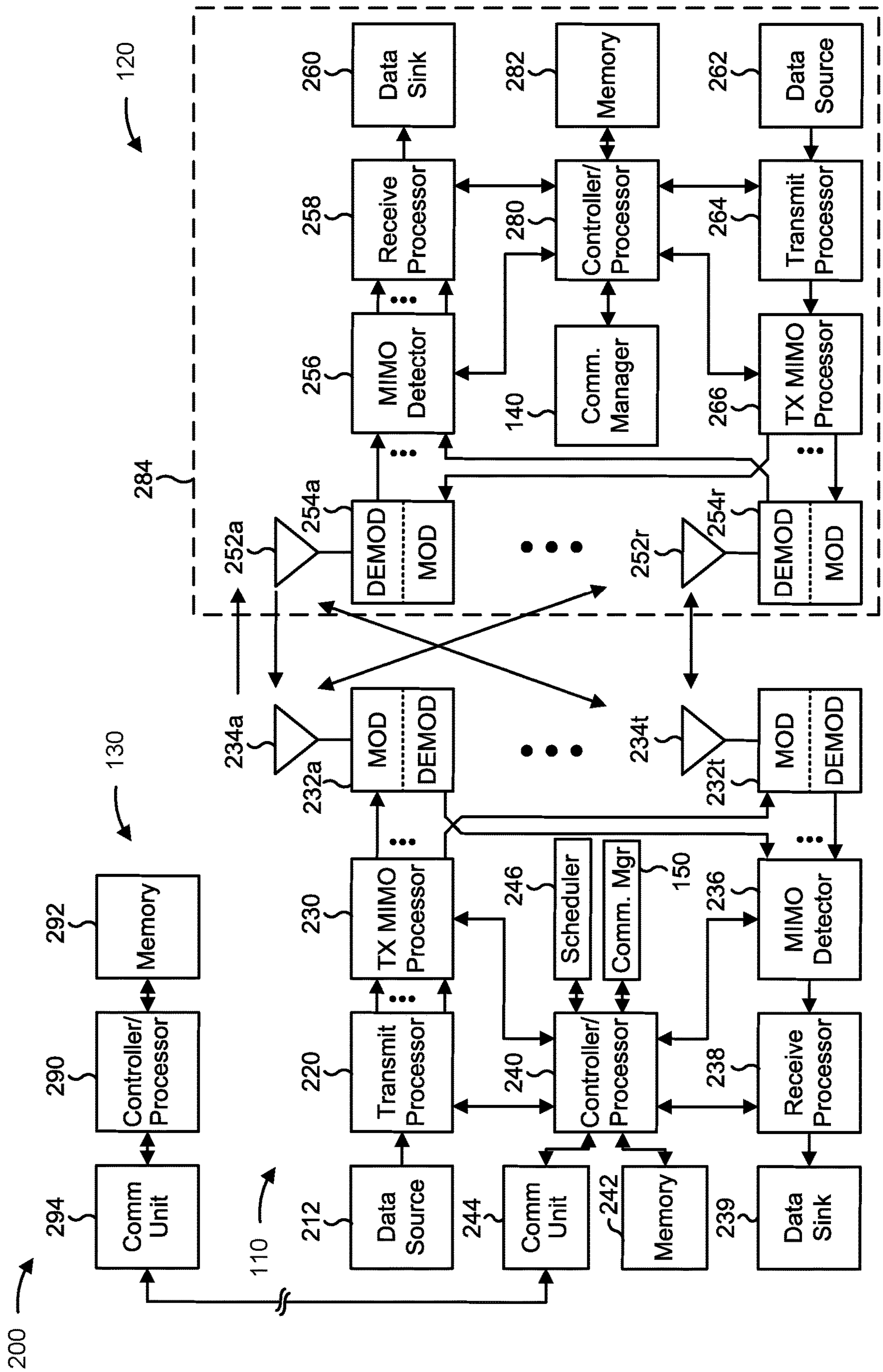

100
Communication Manager 140
Network controller
130
to/from NNs
120a
UE
120e
UE
Macro Cell
120b
110b
NN
102b
Network node(s) (NNs)
110a
Aggregated Architecture
NN
Disaggregated Architecture
NN
NN
NN
102c
110c
NN
120c
102a
110d
Relay NN
120d
UE
Communication Manager 150

FIG. 1



200
110
120
130
284
Data Sink
260
Memory
282
Data Source
262
Receive Processor
258
Controller/ Processor
280
Transmit Processor
264
MIMO Detector
256
Comm. Manager
140
TX MIMO Processor
266
DEMOD
MOD
254a
252a
254r
252r
MOD
DEMOD
234a
232a
234t
232t
TX MIMO Processor
230
Scheduler
246
Comm. Mgr
150
MIMO Detector
236
Transmit Processor
220
Controller/ Processor
240
Receive Processor
238
Data Source
212
Comm Unit
244
Memory
242
Data Sink
239
Memory
292
Controller/ Processor
290
Comm Unit
294

FIG. 2

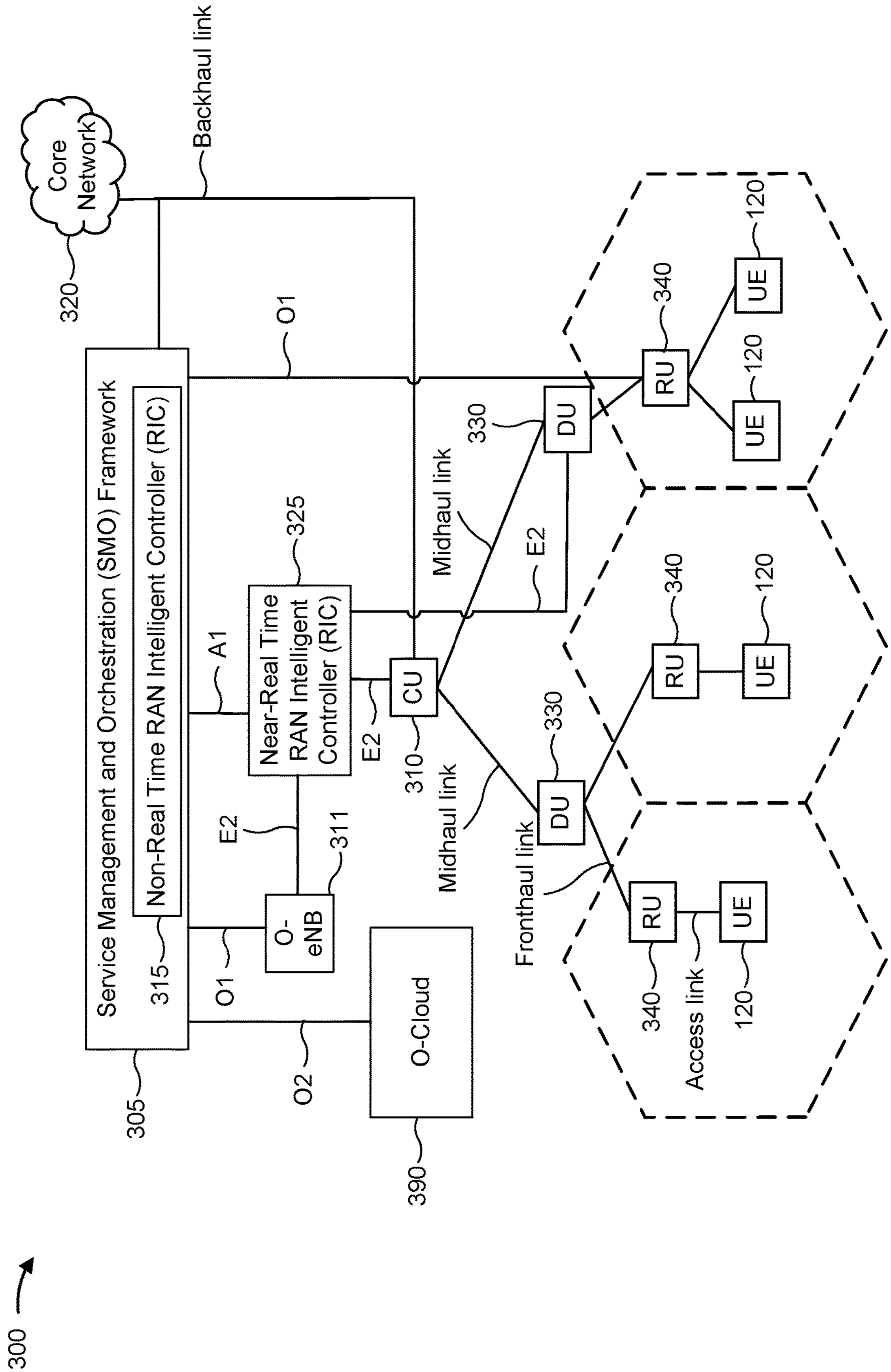
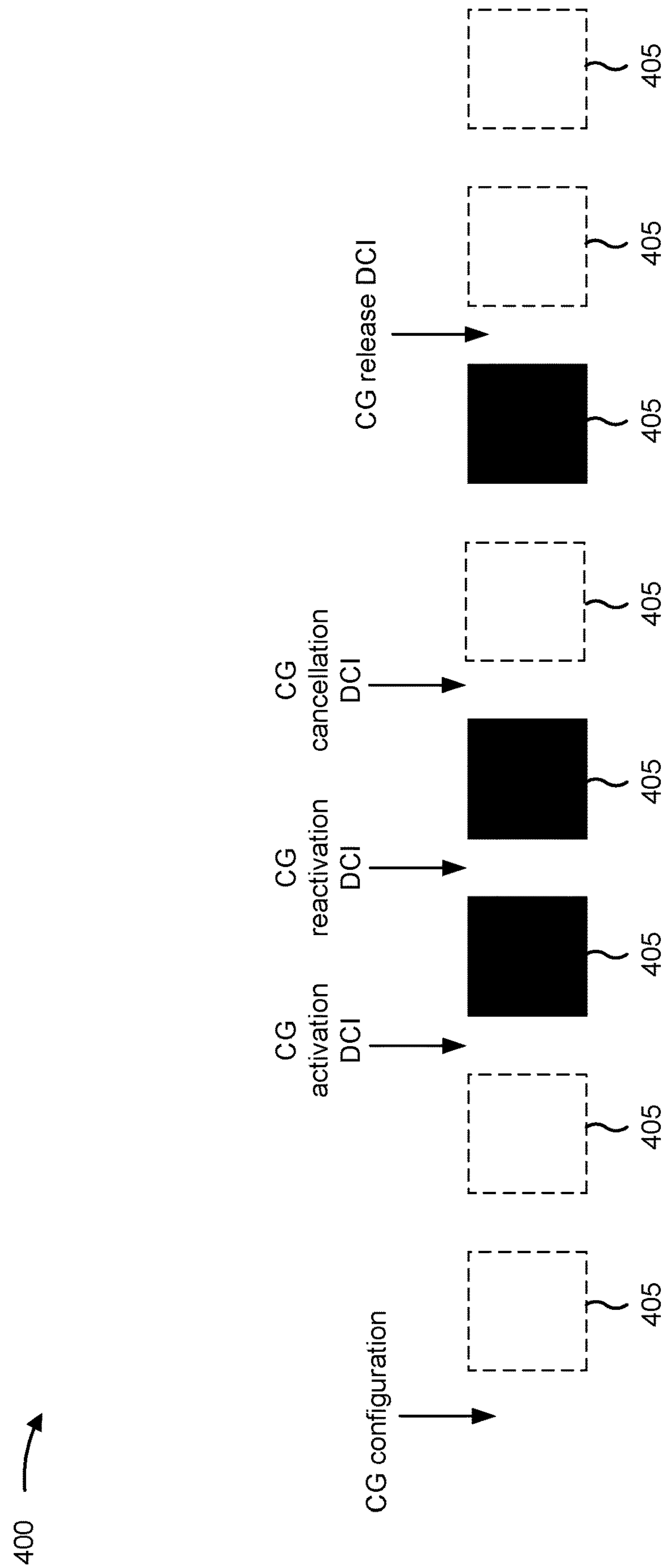

300
Core Network
320
Backhaul link
Service Management and Orchestration (SMO) Framework
Non-Real Time RAN Intelligent Controller (RIC)
315
305
O1
A1
Near-Real Time RAN Intelligent Controller (RIC)
325
E2
O-eNB
311
O1
O2
O-Cloud
390
CU
310
E2
Midhaul link
DU
330
E2
Midhaul link
DU
330
Fronthaul link
RU
340
UE
120
Access link
RU
340
UE
120
RU
340
UE
120
UE
120

FIG. 3



400
CG configuration
405
CG activation DCI
405
CG reactivation DCI
405
CG cancellation DCI
405
405
CG release DCI
405
405

FIG. 4

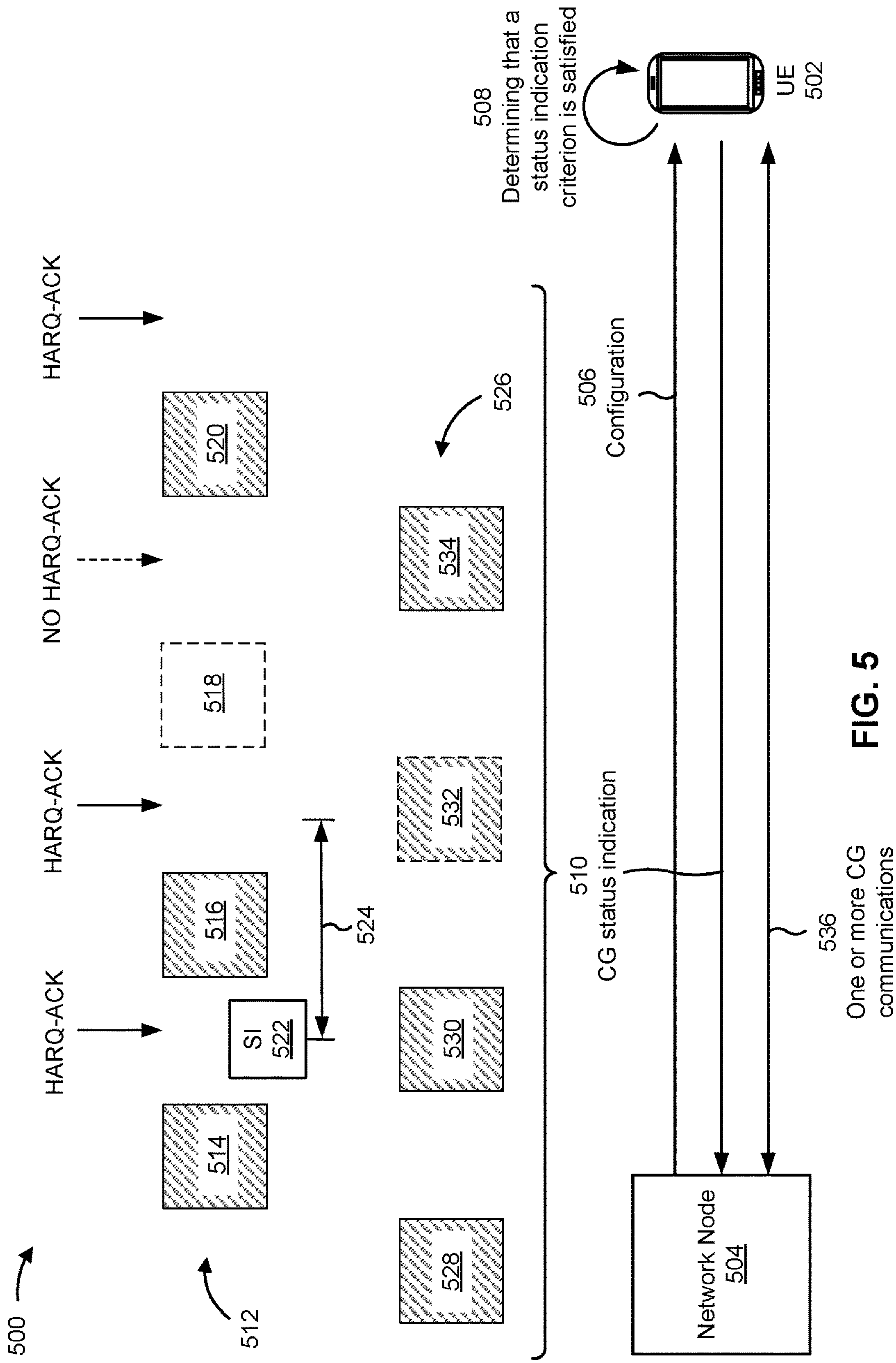

500
512
HARQ-ACK
HARQ-ACK
NO HARQ-ACK
HARQ-ACK
514
516
518
520
SI
522
524
526
528
530
532
534
506
Configuration
508
Determining that a status indication criterion is satisfied
510
CG status indication
536
One or more CG communications
Network Node
504
UE
502

FIG. 5

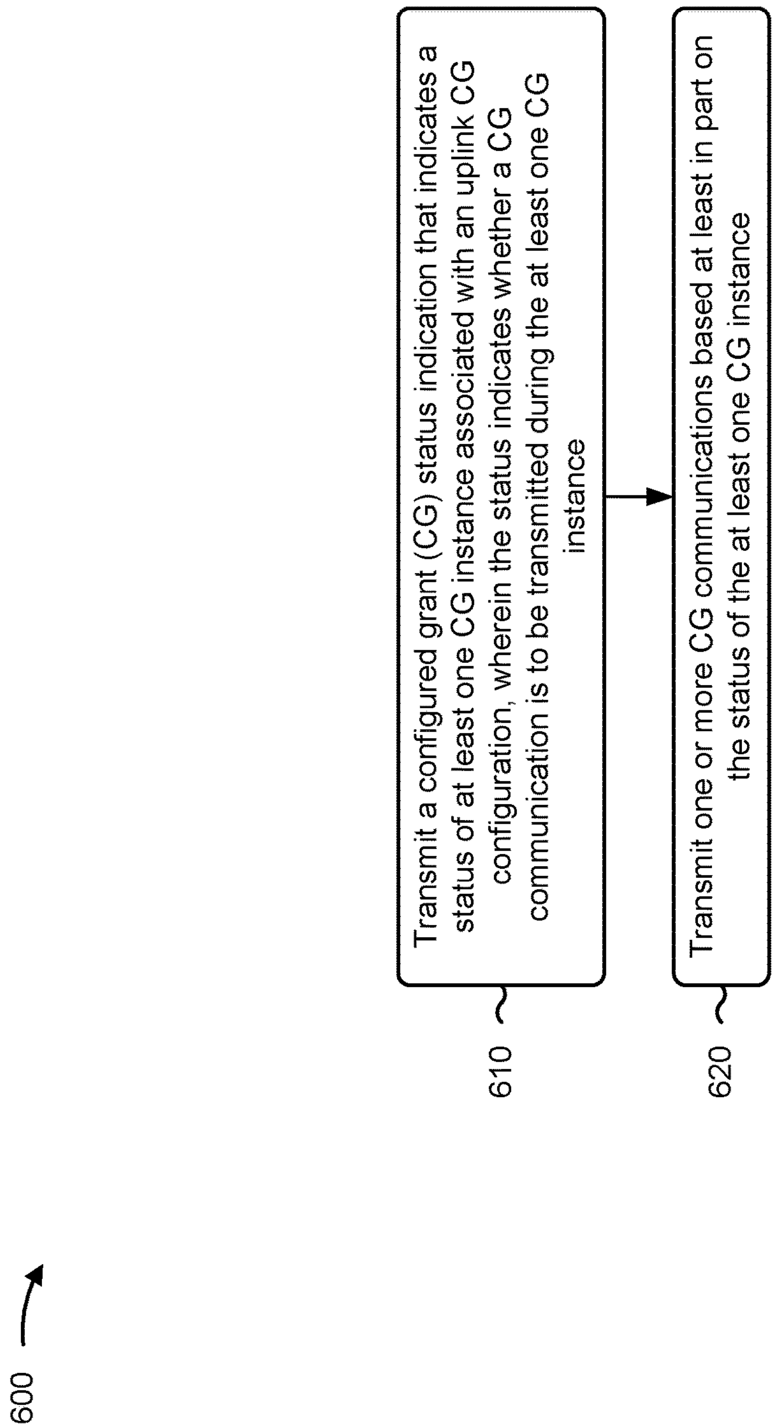

600
610
Transmit a configured grant (CG) status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance
620
Transmit one or more CG communications based at least in part on the status of the at least one CG instance

FIG. 6

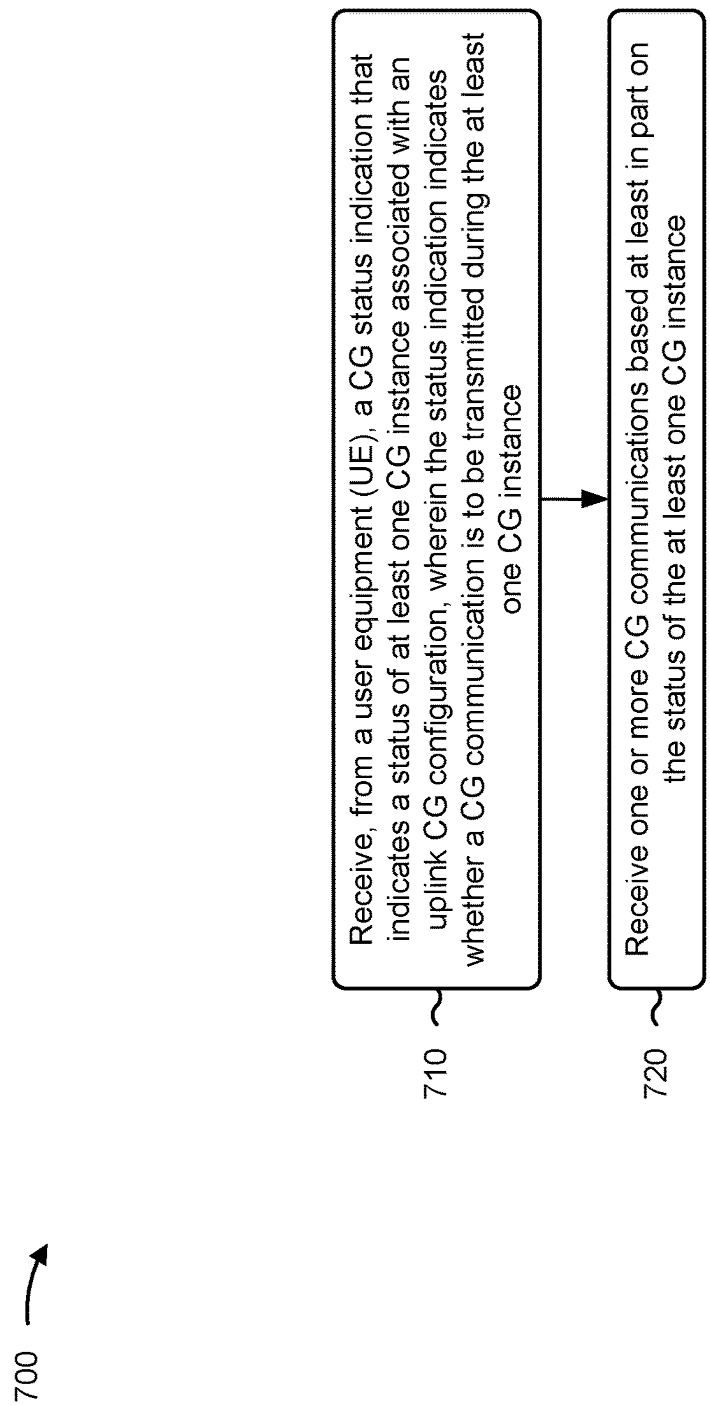

700
710
Receive, from a user equipment (UE), a CG status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indication indicates whether a CG communication is to be transmitted during the at least one CG instance
720
Receive one or more CG communications based at least in part on the status of the at least one CG instance

FIG. 7

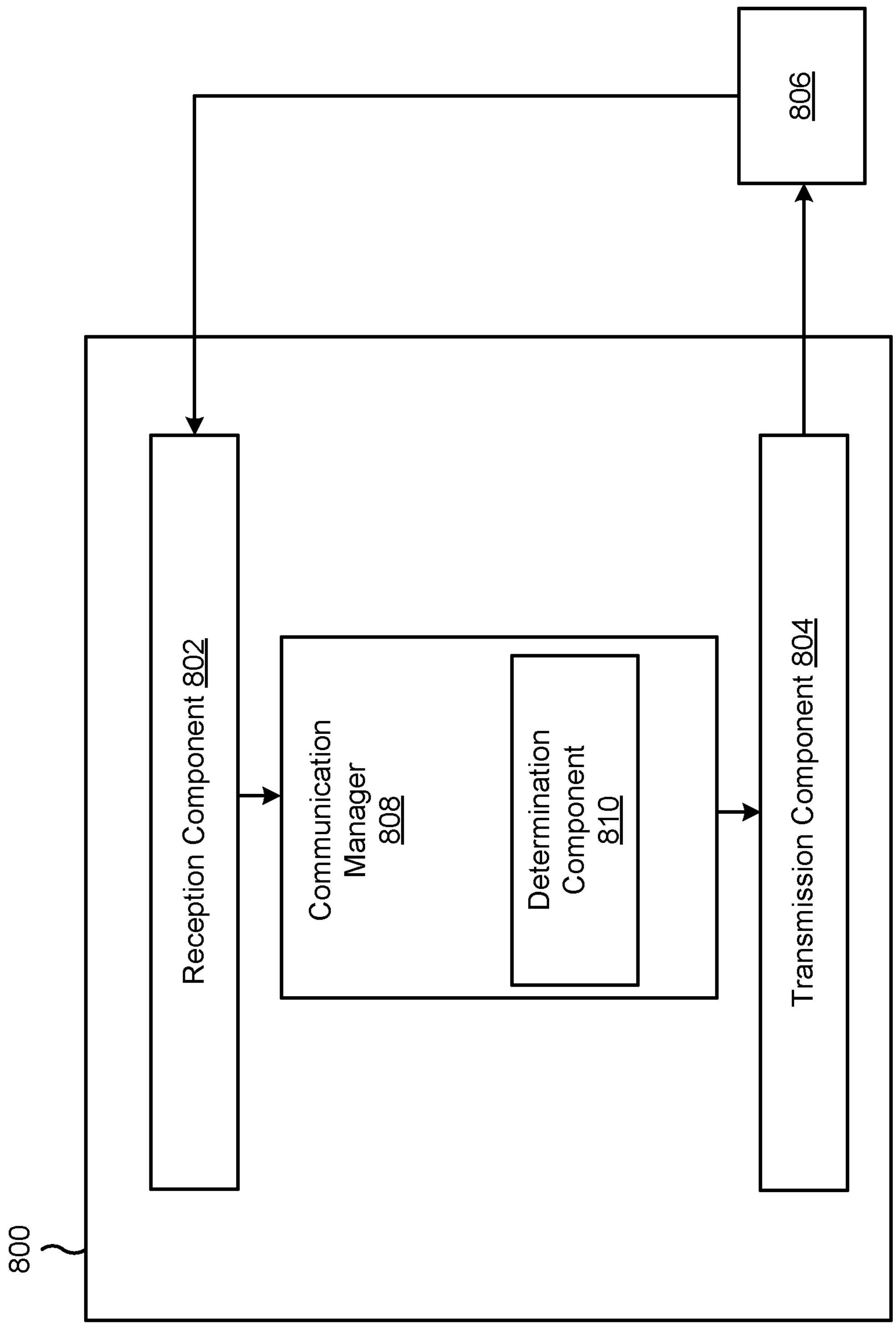

806
800
Reception Component 802
Communication Manager 808
Determination Component 810
Transmission Component 804

FIG. 8

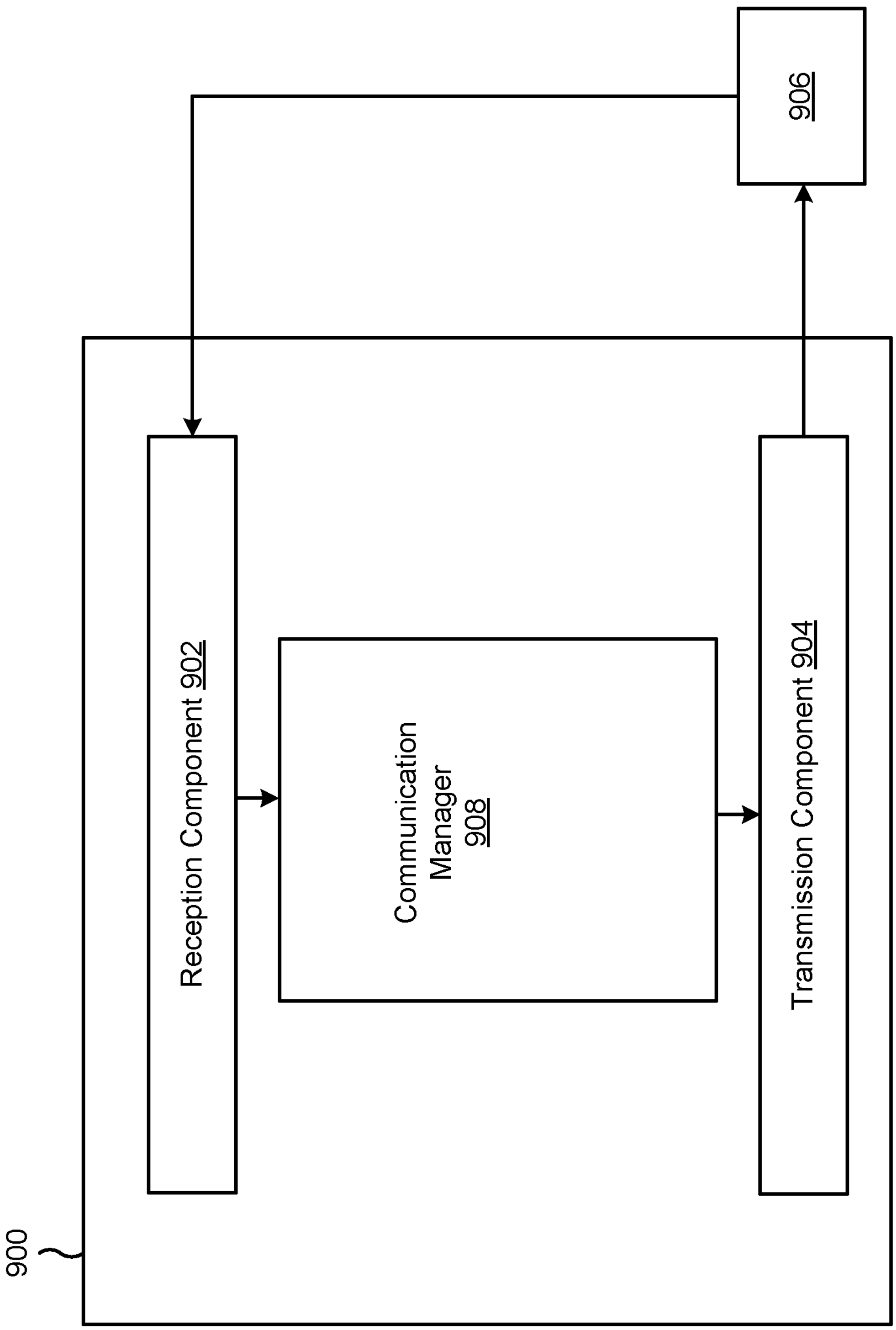

906
Reception Component 902
Communication Manager 908
Transmission Component 904
900

FIG. 9

STATUS INDICATIONS FOR UPLINK CONFIGURED GRANT INSTANCES

FIELD OF THE DISCLOSURE

Aspects of the present disclosure generally relate to wireless communication and to techniques and apparatuses for status indications for uplink configured grant instances.

BACKGROUND

Wireless communication systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, and broadcasts. Typical wireless communication systems may employ multiple-access technologies capable of supporting communication with multiple users by sharing available system resources (e.g., bandwidth, transmit power, or the like). Examples of such multiple-access technologies include code division multiple access (CDMA) systems, time division multiple access (TDMA) systems, frequency division multiple access (FDMA) systems, orthogonal frequency division multiple access (OFDMA) systems, single-carrier frequency division multiple access (SC-FDMA) systems, time division synchronous code division multiple access (TD-SCDMA) systems, and Long Term Evolution (LTE). LTE/LTE-Advanced is a set of enhancements to the Universal Mobile Telecommunications System (UMTS) mobile standard promulgated by the Third Generation Partnership Project (3GPP).

A wireless network may include one or more network nodes that support communication for wireless communication devices, such as a user equipment (UE) or multiple UEs. A UE may communicate with a network node via downlink communications and uplink communications. "Downlink" (or "DL") refers to a communication link from the network node to the UE, and "uplink" (or "UL") refers to a communication link from the UE to the network node. Some wireless networks may support device-to-device communication, such as via a local link (e.g., a sidelink (SL), a wireless local area network (WLAN) link, and/or a wireless personal area network (WPAN) link, among other examples).

The above multiple access technologies have been adopted in various telecommunication standards to provide a common protocol that enables different UEs to communicate on a municipal, national, regional, and/or global level. New Radio (NR), which may be referred to as 5G, is a set of enhancements to the LTE mobile standard promulgated by the 3GPP. NR is designed to better support mobile broadband internet access by improving spectral efficiency, lowering costs, improving services, making use of new spectrum, and better integrating with other open standards using orthogonal frequency division multiplexing (OFDM) with a cyclic prefix (CP) (CP-OFDM) on the downlink, using CP-OFDM and/or single-carrier frequency division multiplexing (SC-FDM) (also known as discrete Fourier transform spread OFDM (DFT-s-OFDM)) on the uplink, as well as supporting beamforming, multiple-input multiple-output (MIMO) antenna technology, and carrier aggregation. As the demand for mobile broadband access continues to increase, further improvements in LTE, NR, and other radio access technologies remain useful.

SUMMARY

Some aspects described herein relate to a user equipment (UE) for wireless communication. The UE may include a memory and one or more processors coupled to the memory. The one or more processors may be configured to transmit a configured grant (CG) status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance. The one or more processors may be configured to transmit one or more CG communications based at least in part on the status of the at least one CG instance.

Some aspects described herein relate to a network node for wireless communication. The network node may include a memory and one or more processors coupled to the memory. The one or more processors may be configured to receive, from a UE, a CG status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance. The one or more processors may be configured to receive one or more CG communications based at least in part on the status of the at least one CG instance.

Some aspects described herein relate to a method of wireless communication performed by a UE. The method may include transmitting a CG status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance. The method may include transmitting one or more CG communications based at least in part on the status of the at least one CG instance.

Some aspects described herein relate to a method of wireless communication performed by a network node. The method may include receiving, from a UE, a CG status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance. The method may include receiving one or more CG communications based at least in part on the status of the at least one CG instance.

Some aspects described herein relate to a non-transitory computer-readable medium that stores a set of instructions for wireless communication by a UE. The set of instructions, when executed by one or more processors of the UE, may cause the UE to transmit a CG status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance. The set of instructions, when executed by one or more processors of the UE, may cause the UE to transmit one or more CG communications based at least in part on the status of the at least one CG instance.

Some aspects described herein relate to a non-transitory computer-readable medium that stores a set of instructions for wireless communication by a network node. The set of instructions, when executed by one or more processors of the network node, may cause the network node to receive, from a UE, a CG status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance. The set of instructions, when executed by one or more processors of the network node, may cause the network node to receive one or more CG communications based at least in part on the status of the at least one CG instance.

Some aspects described herein relate to an apparatus for wireless communication. The apparatus may include means for transmitting a CG status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance. The apparatus may include means for transmitting one or more CG communications based at least in part on the status of the at least one CG instance.

Some aspects described herein relate to an apparatus for wireless communication. The apparatus may include means for receiving, from a UE, a CG status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance. The apparatus may include means for receiving one or more CG communications based at least in part on the status of the at least one CG instance.

Aspects generally include a method, apparatus, system, computer program product, non-transitory computer-readable medium, user equipment, base station, network entity, network node, wireless communication device, and/or processing system as substantially described herein with reference to and as illustrated by the drawings and specification.

The foregoing has outlined rather broadly the features and technical advantages of examples according to the disclosure in order that the detailed description that follows may be better understood. Additional features and advantages will be described hereinafter. The conception and specific examples disclosed may be readily utilized as a basis for modifying or designing other structures for carrying out the same purposes of the present disclosure. Such equivalent constructions do not depart from the scope of the appended claims. Characteristics of the concepts disclosed herein, both their organization and method of operation, together with associated advantages, will be better understood from the following description when considered in connection with the accompanying figures. Each of the figures is provided for the purposes of illustration and description, and not as a definition of the limits of the claims.

While aspects are described in the present disclosure by illustration to some examples, those skilled in the art will understand that such aspects may be implemented in many different arrangements and scenarios. Techniques described herein may be implemented using different platform types, devices, systems, shapes, sizes, and/or packaging arrangements. For example, some aspects may be implemented via integrated chip embodiments or other non-module-component based devices (e.g., end-user devices, vehicles, communication devices, computing devices, industrial equipment, retail/purchasing devices, medical devices, and/or artificial intelligence devices). Aspects may be implemented in chip-level components, modular components, non-modular components, non-chip-level components, device-level components, and/or system-level components. Devices incorporating described aspects and features may include additional components and features for implementation and practice of claimed and described aspects. For example, transmission and reception of wireless signals may include one or more components for analog and digital purposes (e.g., hardware components including antennas, radio frequency (RF) chains, power amplifiers, modulators, buffers, processors, interleavers, adders, and/or summers). It is intended that aspects described herein may be practiced in a wide variety of devices, components, systems, distributed arrangements, and/or end-user devices of varying size, shape, and constitution.

BRIEF DESCRIPTION OF THE DRAWINGS

So that the above-recited features of the present disclosure can be understood in detail, a more particular description, briefly summarized above, may be had by reference to aspects, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only certain typical aspects of this disclosure and are therefore not to be considered limiting of its scope, for the description may admit to other equally effective aspects. The same reference numbers in different drawings may identify the same or similar elements.

FIG. **1** is a diagram illustrating an example of a wireless network, in accordance with the present disclosure.

FIG. **2** is a diagram illustrating an example of a network node in communication with a user equipment (UE) in a wireless network, in accordance with the present disclosure.

FIG. **3** is a diagram illustrating an example disaggregated base station architecture, in accordance with the present disclosure.

FIG. **4** is a diagram illustrating an example of uplink configured grant (CG) communication, in accordance with the present disclosure.

FIG. **5** is a diagram illustrating an example of status indications for uplink CG instances, in accordance with the present disclosure.

FIG. **6** is a diagram illustrating an example process performed, for example, by a UE, in accordance with the present disclosure.

FIG. **7** is a diagram illustrating an example process performed, for example, by a network node, in accordance with the present disclosure.

FIG. **8** is a diagram of an example apparatus for wireless communication, in accordance with the present disclosure.

FIG. **9** is a diagram of an example apparatus for wireless communication, in accordance with the present disclosure.

DETAILED DESCRIPTION

Various aspects of the disclosure are described more fully hereinafter with reference to the accompanying drawings. This disclosure may, however, be embodied in many different forms and should not be construed as limited to any specific structure or function presented throughout this disclosure. Rather, these aspects are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. One skilled in the art should appreciate that the scope of the disclosure is intended to cover any aspect of the disclosure disclosed herein, whether implemented independently of or combined with any other aspect of the disclosure. For example, an apparatus may be implemented or a method may be practiced using any number of the aspects set forth herein. In addition, the scope of the disclosure is intended to cover such an apparatus or method which is practiced using other structure, functionality, or structure and functionality in addition to or other than the various aspects of the disclosure set forth herein. It should be understood that any aspect of the disclosure disclosed herein may be embodied by one or more elements of a claim.

Several aspects of telecommunication systems will now be presented with reference to various apparatuses and techniques. These apparatuses and techniques will be described in the following detailed description and illustrated in the accompanying drawings by various blocks, modules, components, circuits, steps, processes, algorithms, or the like (collectively referred to as "elements"). These elements may be implemented using hardware, software, or combinations thereof. Whether such elements are implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system.

While aspects may be described herein using terminology commonly associated with a 5G or New Radio (NR) radio access technology (RAT), aspects of the present disclosure can be applied to other RATs, such as a 3G RAT, a 4G RAT, and/or a RAT subsequent to 5G (e.g., 6G).

FIG. **1** is a diagram illustrating an example of a wireless network **100**, in accordance with the present disclosure. The wireless network **100** may be or may include elements of a 5G (e.g., NR) network and/or a 4G (e.g., Long Term Evolution (LTE)) network, among other examples. The wireless network **100** may include one or more network nodes **110** (shown as a network node **110***a*, a network node **110***b*, a network node **110***c*, and a network node **110***d*), a user equipment (UE) **120** or multiple UEs **120** (shown as a UE **120***a*, a UE **120***b*, a UE **120***c*, a UE **120***d*, and a UE **120***e*), and/or other entities. A network node **110** is a network node that communicates with UEs **120**. As shown, a network node **110** may include one or more network nodes. For example, a network node **110** may be an aggregated network node, meaning that the aggregated network node is configured to utilize a radio protocol stack that is physically or logically integrated within a single radio access network (RAN) node (e.g., within a single device or unit). As another example, a network node **110** may be a disaggregated network node (sometimes referred to as a disaggregated base station), meaning that the network node **110** is configured to utilize a protocol stack that is physically or logically distributed among two or more nodes (such as one or more central units (CUs), one or more distributed units (DUs), or one or more radio units (RUs)).

In some examples, a network node **110** is or includes a network node that communicates with UEs **120** via a radio access link, such as an RU. In some examples, a network node **110** is or includes a network node that communicates with other network nodes **110** via a fronthaul link or a midhaul link, such as a DU. In some examples, a network node **110** is or includes a network node that communicates with other network nodes **110** via a midhaul link or a core network via a backhaul link, such as a CU. In some examples, a network node **110** (such as an aggregated network node **110** or a disaggregated network node **110**) may include multiple network nodes, such as one or more RUs, one or more CUs, and/or one or more DUs. A network node **110** may include, for example, an NR base station, an LTE base station, a Node B, an eNB (e.g., in 4G), a gNB (e.g., in 5G), an access point, a transmission reception point (TRP), a DU, an RU, a CU, a mobility element of a network, a core network node, a network element, a network equipment, a RAN node, or a combination thereof. In some examples, the network nodes **110** may be interconnected to one another or to one or more other network nodes **110** in the wireless network **100** through various types of fronthaul, midhaul, and/or backhaul interfaces, such as a direct physical connection, an air interface, or a virtual network, using any suitable transport network.

In some examples, a network node **110** may provide communication coverage for a particular geographic area. In the Third Generation Partnership Project (3GPP), the term "cell" can refer to a coverage area of a network node **110** and/or a network node subsystem serving this coverage area, depending on the context in which the term is used. A network node **110** may provide communication coverage for a macro cell, a pico cell, a femto cell, and/or another type of cell. A macro cell may cover a relatively large geographic area (e.g., several kilometers in radius) and may allow unrestricted access by UEs **120** with service subscriptions. A pico cell may cover a relatively small geographic area and may allow unrestricted access by UEs **120** with service subscriptions. A femto cell may cover a relatively small geographic area (e.g., a home) and may allow restricted access by UEs **120** having association with the femto cell (e.g., UEs **120** in a closed subscriber group (CSG)). A network node **110** for a macro cell may be referred to as a macro network node. A network node **110** for a pico cell may be referred to as a pico network node. A network node **110** for a femto cell may be referred to as a femto network node or an in-home network node. In the example shown in FIG. **1**, the network node **110***a* may be a macro network node for a macro cell **102***a*, the network node **110***b* may be a pico network node for a pico cell **102***b*, and the network node **110***c* may be a femto network node for a femto cell **102***c*. A network node may support one or multiple (e.g., three) cells. In some examples, a cell may not necessarily be stationary, and the geographic area of the cell may move according to the location of a network node **110** that is mobile (e.g., a mobile network node).

In some aspects, the term "base station" or "network node" may refer to an aggregated base station, a disaggregated base station, an integrated access and backhaul (IAB) node, a relay node, or one or more components thereof. For example, in some aspects, "base station" or "network node" may refer to a CU, a DU, an RU, a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC), or a Non-Real Time (Non-RT) RIC, or a combination thereof. In some aspects, the term "base station" or "network node" may refer to one device configured to perform one or more functions, such as those described herein in connection with the network node **110**. In some aspects, the term "base station" or "network node" may refer to a plurality of devices configured to perform the one or more functions. For example, in some distributed systems, each of a quantity of different devices (which may be located in the same geographic location or in different geographic locations) may be configured to perform at least a portion of a function, or to duplicate performance of at least a portion of the function, and the term "base station" or "network node" may refer to any one or more of those different devices. In some aspects, the term "base station" or "network node" may refer to one or more virtual base stations or one or more virtual base station functions. For example, in some aspects, two or more base station functions may be instantiated on a single device. In some aspects, the term "base station" or "network node" may refer to one of the base station functions and not another. In this way, a single device may include more than one base station.

The wireless network **100** may include one or more relay stations. A relay station is a network node that can receive a transmission of data from an upstream node (e.g., a network node **110** or a UE **120**) and send a transmission of the data to a downstream node (e.g., a UE **120** or a network node **110**). A relay station may be a UE **120** that can relay transmissions for other UEs **120**. In the example shown in FIG. **1**, the network node **110***d* (e.g., a relay network node) may communicate with the network node **110***a* (e.g., a macro network node) and the UE **120***d* in order to facilitate communication between the network node **110***a* and the UE **120***d*. A network node **110** that relays communications may be referred to as a relay station, a relay base station, a relay network node, a relay node, a relay, or the like.

The wireless network **100** may be a heterogeneous network that includes network nodes **110** of different types, such as macro network nodes, pico network nodes, femto network nodes, relay network nodes, or the like. These different types of network nodes **110** may have different transmit power levels, different coverage areas, and/or different impacts on interference in the wireless network **100**. For example, macro network nodes may have a high transmit power level (e.g., 5 to 40 watts) whereas pico network nodes, femto network nodes, and relay network nodes may have lower transmit power levels (e.g., 0.1 to 2 watts).

A network controller **130** may couple to or communicate with a set of network nodes **110** and may provide coordination and control for these network nodes **110**. The network controller **130** may communicate with the network nodes **110** via a backhaul communication link or a midhaul communication link. The network nodes **110** may communicate with one another directly or indirectly via a wireless or wireline backhaul communication link. In some aspects, the network controller **130** may be a CU or a core network device, or may include a CU or a core network device.

The UEs **120** may be dispersed throughout the wireless network **100**, and each UE **120** may be stationary or mobile. A UE **120** may include, for example, an access terminal, a terminal, a mobile station, and/or a subscriber unit. A UE **120** may be a cellular phone (e.g., a smart phone), a personal digital assistant (PDA), a wireless modem, a wireless communication device, a handheld device, a laptop computer, a cordless phone, a wireless local loop (WLL) station, a tablet, a camera, a gaming device, a netbook, a smartbook, an ultrabook, a medical device, a biometric device, a wearable device (e.g., a smart watch, smart clothing, smart glasses, a smart wristband, smart jewelry (e.g., a smart ring or a smart bracelet)), an entertainment device (e.g., a music device, a video device, and/or a satellite radio), a vehicular component or sensor, a smart meter/sensor, industrial manufacturing equipment, a global positioning system device, a UE function of a network node, and/or any other suitable device that is configured to communicate via a wireless or wired medium.

Some UEs **120** may be considered machine-type communication (MTC) or evolved or enhanced machine-type communication (eMTC) UEs. An MTC UE and/or an eMTC UE may include, for example, a robot, a drone, a remote device, a sensor, a meter, a monitor, and/or a location tag, that may communicate with a network node, another device (e.g., a remote device), or some other entity. Some UEs **120** may be considered Internet-of-Things (IoT) devices, and/or may be implemented as NB-IoT (narrowband IoT) devices. Some UEs **120** may be considered a Customer Premises Equipment. A UE **120** may be included inside a housing that houses components of the UE **120**, such as processor components and/or memory components. In some examples, the processor components and the memory components may be coupled together. For example, the processor components (e.g., one or more processors) and the memory components (e.g., a memory) may be operatively coupled, communicatively coupled, electronically coupled, and/or electrically coupled.

In general, any number of wireless networks **100** may be deployed in a given geographic area. Each wireless network **100** may support a particular RAT and may operate on one or more frequencies. A RAT may be referred to as a radio technology, an air interface, or the like. A frequency may be referred to as a carrier, a frequency channel, or the like. Each frequency may support a single RAT in a given geographic area in order to avoid interference between wireless networks of different RATs. In some cases, NR or 5G RAT networks may be deployed.

In some examples, two or more UEs **120** (e.g., shown as UE **120***a* and UE **120***e*) may communicate directly using one or more sidelink channels (e.g., without using a network node **110** as an intermediary to communicate with one another). For example, the UEs **120** may communicate using peer-to-peer (P2P) communications, device-to-device (D2D) communications, a vehicle-to-everything (V2X) protocol (e.g., which may include a vehicle-to-vehicle (V2V) protocol, a vehicle-to-infrastructure (V2I) protocol, or a vehicle-to-pedestrian (V2P) protocol), and/or a mesh network. In such examples, a UE **120** may perform scheduling operations, resource selection operations, and/or other operations described elsewhere herein as being performed by the network node **110**.

Devices of the wireless network **100** may communicate using the electromagnetic spectrum, which may be subdivided by frequency or wavelength into various classes, bands, channels, or the like. For example, devices of the wireless network **100** may communicate using one or more operating bands. In 5G NR, two initial operating bands have been identified as frequency range designations FR1 (410 MHz-7.125 GHz) and FR2 (24.25 GHz-52.6 GHz). It should be understood that although a portion of FR1 is greater than 6 GHz, FR1 is often referred to (interchangeably) as a "Sub-6 GHz" band in various documents and articles. A similar nomenclature issue sometimes occurs with regard to FR2, which is often referred to (interchangeably) as a "millimeter wave" band in documents and articles, despite being different from the extremely high frequency (EHF) band (30 GHz-300 GHz) which is identified by the International Telecommunications Union (ITU) as a "millimeter wave" band.

The frequencies between FR1 and FR2 are often referred to as mid-band frequencies. Recent 5G NR studies have identified an operating band for these mid-band frequencies as frequency range designation FR3 (7.125 GHz-24.25 GHz). Frequency bands falling within FR3 may inherit FR1 characteristics and/or FR2 characteristics, and thus may effectively extend features of FR1 and/or FR2 into mid-band frequencies. In addition, higher frequency bands are currently being explored to extend 5G NR operation beyond 52.6 GHz. For example, three higher operating bands have been identified as frequency range designations FR4a or FR4-1 (52.6 GHz-71 GHz), FR4 (52.6 GHz-114.25 GHz), and FR5 (114.25 GHz-300 GHz). Each of these higher frequency bands falls within the EHF band.

With the above examples in mind, unless specifically stated otherwise, it should be understood that the term "sub-6 GHz" or the like, if used herein, may broadly represent frequencies that may be less than 6 GHz, may be within FR1, or may include mid-band frequencies. Further, unless specifically stated otherwise, it should be understood that the term "millimeter wave" or the like, if used herein, may broadly represent frequencies that may include mid-band frequencies, may be within FR2, FR4, FR4-a or FR4-1, and/or FR5, or may be within the EHF band. It is contemplated that the frequencies included in these operating bands (e.g., FR1, FR2, FR3, FR4, FR4-a, FR4-1, and/or FR5) may be modified, and techniques described herein are applicable to those modified frequency ranges.

In some aspects, a UE (e.g., the UE **120**) may include a communication manager **140**. As described in more detail elsewhere herein, the communication manager **140** may transmit a configured grant (CG) status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance; and transmit one or more CG communications based at least in part on the status of the at least one CG instance. Additionally, or alternatively, the communication manager **140** may perform one or more other operations described herein.

In some aspects, a network node (e.g., the network node **110**) may include a communication manager **150**. As described in more detail elsewhere herein, the communication manager **150** may receive, from a UE, a CG status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance; and receive one or more CG communications based at least in part on the status of the at least one CG instance. Additionally, or alternatively, the communication manager **150** may perform one or more other operations described herein.

As indicated above, FIG. **1** is provided as an example. Other examples may differ from what is described with regard to FIG. **1**.

FIG. **2** is a diagram illustrating an example **200** of a network node **110** in communication with a UE **120** in a wireless network **100**, in accordance with the present disclosure. The network node **110** may be equipped with a set of antennas **234***a* through **234***t*, such as T antennas (T≥1). The UE **120** may be equipped with a set of antennas **252***a* through **252***r*, such as R antennas (R≥1). The network node **110** of example **200** includes one or more radio frequency components, such as antennas **234** and a modem **254**. In some examples, a network node **110** may include an interface, a communication component, or another component that facilitates communication with the UE **120** or another network node. Some network nodes **110** may not include radio frequency components that facilitate direct communication with the UE **120**, such as one or more CUs, or one or more DUs.

At the network node **110**, a transmit processor **220** may receive data, from a data source **212**, intended for the UE **120** (or a set of UEs **120**). The transmit processor **220** may select one or more modulation and coding schemes (MCSs) for the UE **120** based at least in part on one or more channel quality indicators (CQIs) received from that UE **120**. The network node **110** may process (e.g., encode and modulate) the data for the UE **120** based at least in part on the MCS(s) selected for the UE **120** and may provide data symbols for the UE **120**. The transmit processor **220** may process system information (e.g., for semi-static resource partitioning information (SRPI)) and control information (e.g., CQI requests, grants, and/or upper layer signaling) and provide overhead symbols and control symbols. The transmit processor **220** may generate reference symbols for reference signals (e.g., a cell-specific reference signal (CRS) or a demodulation reference signal (DMRS)) and synchronization signals (e.g., a primary synchronization signal (PSS) or a secondary synchronization signal (SSS)). A transmit (TX) multiple-input multiple-output (MIMO) processor **230** may perform spatial processing (e.g., precoding) on the data symbols, the control symbols, the overhead symbols, and/or the reference symbols, if applicable, and may provide a set of output symbol streams (e.g., T output symbol streams) to a corresponding set of modems **232** (e.g., T modems), shown as modems **232***a* through **232***t*. For example, each output symbol stream may be provided to a modulator component (shown as MOD) of a modem **232**. Each modem **232** may use a respective modulator component to process a respective output symbol stream (e.g., for OFDM) to obtain an output sample stream. Each modem **232** may further use a respective modulator component to process (e.g., convert to analog, amplify, filter, and/or upconvert) the output sample stream to obtain a downlink signal. The modems **232***a* through **232***t* may transmit a set of downlink signals (e.g., T downlink signals) via a corresponding set of antennas **234** (e.g., T antennas), shown as antennas **234***a* through **234***t*.

At the UE **120**, a set of antennas **252** (shown as antennas **252***a* through **252***r*) may receive the downlink signals from the network node **110** and/or other network nodes **110** and may provide a set of received signals (e.g., R received signals) to a set of modems **254** (e.g., R modems), shown as modems **254***a* through **254***r*. For example, each received signal may be provided to a demodulator component (shown as DEMOD) of a modem **254**. Each modem **254** may use a respective demodulator component to condition (e.g., filter, amplify, downconvert, and/or digitize) a received signal to obtain input samples. Each modem **254** may use a demodulator component to further process the input samples (e.g., for OFDM) to obtain received symbols. A MIMO detector **256** may obtain received symbols from the modems **254**, may perform MIMO detection on the received symbols if applicable, and may provide detected symbols. A receive processor **258** may process (e.g., demodulate and decode) the detected symbols, may provide decoded data for the UE **120** to a data sink **260**, and may provide decoded control information and system information to a controller/processor **280**. The term "controller/processor" may refer to one or more controllers, one or more processors, or a combination thereof. A channel processor may determine a reference signal received power (RSRP) parameter, a received signal strength indicator (RSSI) parameter, a reference signal received quality (RSRQ) parameter, and/or a CQI parameter, among other examples. In some examples, one or more components of the UE **120** may be included in a housing **284**.

The network controller **130** may include a communication unit **294**, a controller/processor **290**, and a memory **292**. The network controller **130** may include, for example, one or more devices in a core network. The network controller **130** may communicate with the network node **110** via the communication unit **294**.

One or more antennas (e.g., antennas **234***a* through **234***t* and/or antennas **252***a* through **252***r*) may include, or may be included within, one or more antenna panels, one or more antenna groups, one or more sets of antenna elements, and/or one or more antenna arrays, among other examples. An antenna panel, an antenna group, a set of antenna elements, and/or an antenna array may include one or more antenna elements (within a single housing or multiple housings), a set of coplanar antenna elements, a set of non-coplanar antenna elements, and/or one or more antenna elements coupled to one or more transmission and/or reception components, such as one or more components of FIG. **2**.

On the uplink, at the UE **120**, a transmit processor **264** may receive and process data from a data source **262** and control information (e.g., for reports that include RSRP, RSSI, RSRQ, and/or CQI) from the controller/processor **280**. The transmit processor **264** may generate reference symbols for one or more reference signals. The symbols from the transmit processor **264** may be precoded by a TX MIMO processor **266** if applicable, further processed by the modems **254** (e.g., for DFT-s-OFDM or CP-OFDM), and transmitted to the network node **110**. In some examples, the modem **254** of the UE **120** may include a modulator and a demodulator. In some examples, the UE **120** includes a transceiver. The transceiver may include any combination of the antenna(s) **252**, the modem(s) **254**, the MIMO detector **256**, the receive processor **258**, the transmit processor **264**, and/or the TX MIMO processor **266**. The transceiver may be used by a processor (e.g., the controller/processor **280**) and the memory **282** to perform aspects of any of the methods described herein (e.g., with reference to FIGS. **5-9**).

At the network node **110**, the uplink signals from UE **120** and/or other UEs may be received by the antennas **234**, processed by the modem **232** (e.g., a demodulator component, shown as DEMOD, of the modem **232**), detected by a MIMO detector **236** if applicable, and further processed by a receive processor **238** to obtain decoded data and control information sent by the UE **120**. The receive processor **238** may provide the decoded data to a data sink **239** and provide the decoded control information to the controller/processor **240**. The network node **110** may include a communication unit **244** and may communicate with the network controller **130** via the communication unit **244**. The network node **110** may include a scheduler **246** to schedule one or more UEs **120** for downlink and/or uplink communications. In some examples, the modem **232** of the network node **110** may include a modulator and a demodulator. In some examples, the network node **110** includes a transceiver. The transceiver may include any combination of the antenna(s) **234**, the modem(s) **232**, the MIMO detector **236**, the receive processor **238**, the transmit processor **220**, and/or the TX MIMO processor **230**. The transceiver may be used by a processor (e.g., the controller/processor **240**) and the memory **242** to perform aspects of any of the methods described herein (e.g., with reference to FIGS. **5-9**).

The controller/processor **240** of the network node **110**, the controller/processor **280** of the UE **120**, and/or any other component(s) of FIG. **2** may perform one or more techniques associated with status indications for uplink CG instances, as described in more detail elsewhere herein. For example, the controller/processor **240** of the network node **110**, the controller/processor **280** of the UE **120**, and/or any other component(s) of FIG. **2** may perform or direct operations of, for example, process **600** of FIG. **6**, process **700** of FIG. **7**, and/or other processes as described herein. The memory **242** and the memory **282** may store data and program codes for the network node **110** and the UE **120**, respectively. In some examples, the memory **242** and/or the memory **282** may include a non-transitory computer-readable medium storing one or more instructions (e.g., code and/or program code) for wireless communication. For example, the one or more instructions, when executed (e.g., directly, or after compiling, converting, and/or interpreting) by one or more processors of the network node **110** and/or the UE **120**, may cause the one or more processors, the UE **120**, and/or the network node **110** to perform or direct operations of, for example, process **600** of FIG. **6**, process **700** of FIG. **7**, and/or other processes as described herein. In some examples, executing instructions may include running the instructions, converting the instructions, compiling the instructions, and/or interpreting the instructions, among other examples.

In some aspects, a UE (e.g., the UE **120**) includes means for transmitting a CG status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance; and/or means for transmitting one or more CG communications based at least in part on the status of the at least one CG instance. The means for the UE to perform operations described herein may include, for example, one or more of communication manager **140**, antenna **252**, modem **254**, MIMO detector **256**, receive processor **258**, transmit processor **264**, TX MIMO processor **266**, controller/processor **280**, or memory **282**.

In some aspects, the network node includes means for receiving, from a UE, a CG status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance; and/or means for receiving one or more CG communications based at least in part on the status of the at least one CG instance. In some aspects, the means for the network node to perform operations described herein may include, for example, one or more of communication manager **150**, transmit processor **220**, TX MIMO processor **230**, modem **232**, antenna **234**, MIMO detector **236**, receive processor **238**, controller/processor **240**, memory **242**, or scheduler **246**.

While blocks in FIG. **2** are illustrated as distinct components, the functions described above with respect to the blocks may be implemented in a single hardware, software, or combination component or in various combinations of components. For example, the functions described with respect to the transmit processor **264**, the receive processor **258**, and/or the TX MIMO processor **266** may be performed by or under the control of the controller/processor **280**.

As indicated above, FIG. **2** is provided as an example. Other examples may differ from what is described with regard to FIG. **2**.

Deployment of communication systems, such as 5GNR systems, may be arranged in multiple manners with various components or constituent parts. In a 5G NR system, or network, a network node, a network entity, a mobility element of a network, a RAN node, a core network node, a network element, a base station, or a network equipment may be implemented in an aggregated or disaggregated architecture. For example, a base station (such as a Node B (NB), an evolved NB (eNB), an NR BS, a 5G NB, an access point (AP), a TRP, or a cell, among other examples), or one or more units (or one or more components) performing base station functionality, may be implemented as an aggregated base station (also known as a standalone base station or a monolithic base station) or a disaggregated base station. "Network entity" or "network node" may refer to a disaggregated base station, or to one or more units of a disaggregated base station (such as one or more CUs, one or more DUs, one or more RUs, or a combination thereof).

An aggregated base station (e.g., an aggregated network node) may be configured to utilize a radio protocol stack that is physically or logically integrated within a single RAN node (e.g., within a single device or unit). A disaggregated base station (e.g., a disaggregated network node) may be configured to utilize a protocol stack that is physically or logically distributed among two or more units (such as one or more CUs, one or more DUs, or one or more RUs). In some examples, a CU may be implemented within a network node, and one or more DUs may be co-located with the CU, or alternatively, may be geographically or virtually distributed throughout one or multiple other network nodes. The DUs may be implemented to communicate with one or more RUs. Each of the CU, DU, and RU also can be implemented as virtual units, such as a virtual central unit (VCU), a virtual distributed unit (VDU), or a virtual radio unit (VRU), among other examples.

Base station-type operation or network design may consider aggregation characteristics of base station functionality. For example, disaggregated base stations may be utilized in an IAB network, an open radio access network (O-RAN (such as the network configuration sponsored by the O-RAN Alliance)), or a virtualized radio access network (vRAN, also known as a cloud radio access network (C-RAN)) to facilitate scaling of communication systems by separating base station functionality into one or more units that can be individually deployed. A disaggregated base station may include functionality implemented across two or more units at various physical locations, as well as functionality implemented for at least one unit virtually, which can enable flexibility in network design. The various units of the disaggregated base station can be configured for wired or wireless communication with at least one other unit of the disaggregated base station.

FIG. **3** is a diagram illustrating an example disaggregated base station architecture **300**, in accordance with the present disclosure. The disaggregated base station architecture **300** may include a CU **310** that can communicate directly with a core network **320** via a backhaul link, or indirectly with the core network **320** through one or more disaggregated control units (such as a Near-RT RIC **325** via an E2 link, or a Non-RT RIC **315** associated with a Service Management and Orchestration (SMO) Framework **305**, or both). A CU **310** may communicate with one or more DUs **330** via respective midhaul links, such as through F1 interfaces. Each of the DUs **330** may communicate with one or more RUs **340** via respective fronthaul links. Each of the RUs **340** may communicate with one or more UEs **120** via respective radio frequency (RF) access links. In some implementations, a UE **120** may be simultaneously served by multiple RUs **340**.

Each of the units, including the CUs **310**, the DUs **330**, the RUs **340**, as well as the Near-RT RICs **325**, the Non-RT RICs **315**, and the SMO Framework **305**, may include one or more interfaces or be coupled with one or more interfaces configured to receive or transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to one or multiple communication interfaces of the respective unit, can be configured to communicate with one or more of the other units via the transmission medium. In some examples, each of the units can include a wired interface, configured to receive or transmit signals over a wired transmission medium to one or more of the other units, and a wireless interface, which may include a receiver, a transmitter or transceiver (such as an RF transceiver), configured to receive or transmit signals, or both, over a wireless transmission medium to one or more of the other units.

In some aspects, the CU **310** may host one or more higher layer control functions. Such control functions can include radio resource control (RRC) functions, packet data convergence protocol (PDCP) functions, or service data adaptation protocol (SDAP) functions, among other examples. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU **310**. The CU **310** may be configured to handle user plane functionality (for example, Central Unit-User Plane (CU-UP) functionality), control plane functionality (for example, Central Unit-Control Plane (CU-CP) functionality), or a combination thereof. In some implementations, the CU **310** can be logically split into one or more CU-UP units and one or more CU-CP units. A CU-UP unit can communicate bidirectionally with a CU-CP unit via an interface, such as the E1 interface when implemented in an O-RAN configuration. The CU **310** can be implemented to communicate with a DU **330**, as necessary, for network control and signaling.

Each DU **330** may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs **340**. In some aspects, the DU **330** may host one or more of a radio link control (RLC) layer, a MAC layer, and one or more high physical (PHY) layers depending, at least in part, on a functional split, such as a functional split defined by the 3GPP. In some aspects, the one or more high PHY layers may be implemented by one or more modules for forward error correction (FEC) encoding and decoding, scrambling, and modulation and demodulation, among other examples. In some aspects, the DU **330** may further host one or more low PHY layers, such as implemented by one or more modules for a fast Fourier transform (FFT), an inverse FFT (iFFT), digital beamforming, or physical random access channel (PRACH) extraction and filtering, among other examples. Each layer (which also may be referred to as a module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU **330**, or with the control functions hosted by the CU **310**.

Each RU **340** may implement lower-layer functionality. In some deployments, an RU **340**, controlled by a DU **330**, may correspond to a logical node that hosts RF processing functions or low-PHY layer functions, such as performing an FFT, performing an iFFT, digital beamforming, or PRACH extraction and filtering, among other examples, based on a functional split (for example, a functional split defined by the 3GPP), such as a lower layer functional split. In such an architecture, each RU **340** can be operated to handle over the air (OTA) communication with one or more UEs **120**. In some implementations, real-time and non-real-time aspects of control and user plane communication with the RU(s) **340** can be controlled by the corresponding DU **330**. In some scenarios, this configuration can enable each DU **330** and the CU **310** to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.

The SMO Framework **305** may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework **305** may be configured to support the deployment of dedicated physical resources for RAN coverage requirements, which may be managed via an operations and maintenance interface (such as an O1 interface).

For virtualized network elements, the SMO Framework **305** may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) platform **390**) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements can include, but are not limited to, CUs **310**, DUs **330**, RUs **340**, non-RT RICs **315**, and Near-RT RICs **325**. In some implementations, the SMO Framework **305** can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) **311**, via an O1 interface. Additionally, in some implementations, the SMO Framework **305** can communicate directly with each of one or more RUs **340** via a respective O1 interface. The SMO Framework **305** also may include a Non-RT RIC **315** configured to support functionality of the SMO Framework **305**.

The Non-RT RIC **315** may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, Artificial Intelligence/Machine Learning (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC **325**. The Non-RT

RIC **315** may be coupled to or communicate with (such as via an A1 interface) the Near-RT RIC **325**. The Near-RT RIC **325** may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs **310**, one or more DUs **330**, or both, as well as an O-eNB, with the Near-RT RIC **325**.

In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC **325**, the Non-RT RIC **315** may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC **325** and may be received at the SMO Framework **305** or the Non-RT RIC **315** from non-network data sources or from network functions. In some examples, the Non-RT RIC **315** or the Near-RT RIC **325** may be configured to tune RAN behavior or performance. For example, the Non-RT RIC **315** may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework **305** (such as reconfiguration via an O1 interface) or via creation of RAN management policies (such as A1 interface policies).

As indicated above, FIG. **3** is provided as an example. Other examples may differ from what is described with regard to FIG. **3**.

FIG. **4** is a diagram illustrating an example **400** of uplink CG communication, in accordance with the present disclosure. CG communications may include periodic uplink communications that are configured for a UE, such that the network node does not need to send separate downlink control information (DCI) to schedule each uplink communication, thereby conserving signaling overhead.

As shown in example **400**, a UE may be configured with a CG configuration for CG communications. For example, the UE may receive the CG configuration via a radio resource control (RRC) message transmitted by a network node. The CG configuration may indicate a resource allocation associated with CG uplink communications (e.g., in a time domain, frequency domain, spatial domain, and/or code domain) and a periodicity at which the resource allocation is repeated, resulting in periodically reoccurring scheduled CG instances **405** for the UE. A CG instance **405** can include a CG occasion, a set of CG occasions, or a portion of a CG occasion. In some examples, the CG configuration may identify a resource pool or multiple resource pools that are available to the UE for an uplink transmission. The CG configuration may configure contention-free CG communications (e.g., where resources are dedicated for the UE to transmit uplink communications) or contention-based CG communications (e.g., where the UE contends for access to a channel in the configured resource allocation, such as by using a channel access procedure or a channel sensing procedure).

The network node may transmit CG activation DCI to the UE to activate the CG configuration for the UE. The network node may indicate, in the CG activation DCI, communication parameters, such as a modulation and coding scheme (MCS), a resource block (RB) allocation, and/or antenna ports, for the CG physical uplink shared channel (PUSCH) communications to be transmitted in the scheduled CG instances **405**. The UE may begin transmitting in the CG instances **405** based at least in part on receiving the CG activation DCI. For example, beginning with a next scheduled CG instance **405** subsequent to receiving the CG activation DCI, the UE may transmit a PUSCH communication in the scheduled CG instances **405** using the communication parameters indicated in the CG activation DCI. The UE may refrain from transmitting in configured CG instances **405** prior to receiving the CG activation DCI.

The network node may transmit CG reactivation DCI to the UE to change the communication parameters for the CG PUSCH communications. Based at least in part on receiving the CG reactivation DCI, the UE may begin transmitting in the scheduled CG instances **405** using the communication parameters indicated in the CG reactivation DCI. For example, beginning with a next scheduled CG instance **405** subsequent to receiving the CG reactivation DCI, the UE may transmit PUSCH communications in the scheduled CG instances **405** based at least in part on the communication parameters indicated in the CG reactivation DCI.

In some cases, such as when the network node needs to override a scheduled CG communication for a higher priority communication, the network node may transmit CG cancellation DCI to the UE to temporarily cancel or deactivate one or more subsequent CG instances **405** for the UE. The CG cancellation DCI may deactivate only a subsequent one CG instance **405** or a subsequent N CG instances **405** (where Nis an integer). CG instances **405** after the one or more (e.g., N) CG instances **405** subsequent to the CG cancellation DCI may remain activated. Based at least in part on receiving the CG cancellation DCI, the UE may refrain from transmitting in the one or more (e.g., N) CG instances **405** subsequent to receiving the CG cancellation DCI. As shown in example **400**, the CG cancellation DCI cancels one subsequent CG instance **405** for the UE. After the CG instances **405** (or N CG occasions) subsequent to receiving the CG cancellation DCI, the UE may automatically resume transmission in the scheduled CG instances **405**.

The network node may transmit CG release DCI to the UE to deactivate the CG configuration for the UE. The UE may stop transmitting in the scheduled CG instances **405** based at least in part on receiving the CG release DCI. For example, the UE may refrain from transmitting in any scheduled CG instances **405** until another CG activation DCI is received from the network node. Whereas the CG cancellation DCI may deactivate only a subsequent one CG instance **405** or a subsequent N CG instances **405**, the CG release DCI deactivates all subsequent CG instances **405** for a given CG configuration for the UE until the given CG configuration is activated again by a new CG activation DCI.

In some cases, uplink CG can be configured according to two different CG types. In one type, DCI activation is needed for initiation of the CG transmissions, and in the other type, DCI activation is not needed. Generally, a UE configured with CG transmits in an uplink CG instance only if the UE has data to transmit. In some cases, the UE can determine that there is no data to transmit during a CG instance and, as a result, will not transmit during the CG instance. In some cases, a UE can determine not to transmit data that is ready for transmission during a CG instance. When a UE refrains from transmitting during a configured CG instance, the UE can be said to "skip" the configured CG instance. Similarly, the practice of refraining from transmitting during a configured CG instance can be referred to as "skipping."

However, a network node that configures the CG can be unaware of whether the UE has data to transmit or not, and so the network node typically performs decoding during the configured CG instances regardless. As a result, the network node can waste processing power and time decoding CG instances when there has not been a transmission. Moreover, the network node sometimes can still transmit a hybrid automatic response if a hybrid automatic repeat request (HARQ)-acknowledgement (ACK) (HARQ-ACK) is configured.

In some cases, the network node can determine that the network node is to transition into a sleep mode to save energy, however, the existence of a CG instance may either prevent the network node from transitioning to the sleep mode or may cause the network node to divide a deep sleep mode into two lighter sleep modes around the CG instance. A deep sleep mode is a sleep mode in which more processes are deactivated than are deactivated in a lighter (or "light") sleep mode. In this way, energy savings can be sacrificed due to the network node not being aware of whether the UE will transmit during a particular CG instance. In some cases, the network node can configure uplink CG with a high periodicity to save energy, but higher periodicity can result in higher UE latency.

Some aspects of the techniques and apparatuses described herein provide support for status indications for uplink CG instances. In some aspects, the techniques and apparatuses described herein may apply to access link communications and/or sidelink communications. A CG communication, for example, may include an uplink CG communication, a downlink semi-persistent scheduling (SPS) communication, a sidelink CG communication, or a sidelink SPS communication. In some aspects, for example, a UE may transmit a CG status indication that indicates a status of at least one CG instance associated with an uplink CG configuration. The status indication may indicate whether a CG communication is to be transmitted during the at least one CG instance. The UE may transmit one or more CG communications based at least in part on the status of the at least one CG instance. For example, the CG status indication may include a skipping indication that indicates at least one skipped CG instance during which a CG communication is not to be transmitted. In this way, some aspects may facilitate making a network node aware of whether a UE is going to transmit during a specified CG instance. The network node may refrain from decoding a CG instance that is going to be skipped and may transition to a sleep state during the CG instance. As a result, some aspects may facilitate improving communication efficiency, saving of energy at the network node, and positively impacting network performance.

As indicated above, FIG. **4** is provided as an example. Other examples may differ from what is described with respect to FIG. **4**.

FIG. **5** is a diagram illustrating an example **500** of status indications for uplink CG instances, in accordance with the present disclosure. As shown in FIG. **5**, a network UE **502** and a network node **504** may communicate with one another. In some aspects, the UE **502** may be, be similar to, include, or be included in, the UE **120** depicted in FIGS. **1** and **2**. In some aspects, the network node **504** may be, be similar to, include, or be included in, the network node **110** depicted in FIGS. **1** and **2**.

As shown by reference number **506**, the UE **502** may transmit, and the network node **504** may receive, a configuration. For example, the configuration may be transmitted using an RRC message and may include an uplink CG configuration. In some aspects, the configuration may indicate a maximum quantity of skipped CG instances. The maximum quantity of skipped CG instances may be based at least in part on a periodicity corresponding to the CG configuration. In some aspects, the maximum quantity of skipped CG instances may be specified in a wireless communication standard. For example, a memory of the UE may include an indication of a relation between the periodicity and the maximum quantity of skipped CG instances. In some aspects, the configuration may indicate a mapping between the periodicity and the maximum quantity of skipped CG instances.

In some aspects, the configuration may include a buffer status report (BSR) configuration. The BSR configuration may indicate a BSR threshold such that the UE **502** does not transmit a CG communication associated with at least one CG instance, based at least in part on a buffer status associated with the at least one CG instance not satisfying the BSR threshold. In this way, the UE **502** may be configured to transmit data in a CG instance only when a specified amount of data is in the buffer to transmit.

As shown by reference number **508**, the UE **502** may determine that a status indication criterion is satisfied. In some aspects, the status indication criterion may be satisfied based at least in part on a data size corresponding to an uplink buffer satisfying a data size threshold. The data size threshold may include, for example, the BSR threshold. In some aspects, the status indication criterion may be satisfied based at least in part on an output of a machine learning model associated with incoming data traffic. For example, a machine learning model may be configured on the UE **502** to predict CG occasions that may be skipped based on communication needs of the UE **502** and/or data transmission patterns of the UE **502**, among other examples.

As shown by reference number **510**, the UE **502** may transmit, and the network node **504** may receive, a CG status indication that indicates a status of at least one CG instance associated with the uplink CG configuration. The status indication may indicate whether a CG communication is to be transmitted during the at least one CG instance. In some aspects, the UE **502** may transmit the CG status indication based at least in part on determining that the status indication criterion is satisfied.

In some aspects, the CG status indication may indicate a quantity of CG instances of the at least one CG instance. In some aspects, the CG status indication may include a skipping indication that indicates at least one skipped CG instance, of a plurality of CG instances that includes the at least one CG instance, during which a CG communication is not to be transmitted. In some aspects, the CG status indication may indicate a quantity of skipped CG instances, of the at least one skipped CG instance. The quantity of skipped CG instances may be no greater than the maximum quantity of skipped CG instances.

In some aspects, the CG status indication may correspond to a specified time period. As shown in FIG. **5**, for example, the UE **502** may be configured with a CG configuration **512** associated with CG instances **514**, **516**, **518**, and **520**. The UE **502** may transmit a CG communication during the CG instance **514** (as indicated by the hatch pattern in the illustrated box) and may transmit a CG status indication **522** (shown as "SI") after transmitting the CG communication during the CG instance **514**. In some aspects, a specified time period **524** may begin at the time of the transmission of the CG status indication **522**, at a time instance of reception, by the network node **504**, of the CG status indication **522**, or at a time instance associated with an offset with respect to transmission of the CG status indication **522** or reception, by the network node **504**, of the CG status indication **522**. In some aspects, the UE **502** may transmit a CG communication during the CG instance **516** since the CG instance **516** falls within the specified time period **524**, but may not transmit a CG communication during the CG instance **518** since the CG instance **518** occurs after the end of the specified time period **524**.

In some aspects, the UE **502** may transmit the CG status indication by transmitting a CG communication, associated with a last CG instance before the at least one CG instance, where the CG communication includes the CG status indication. The last CG instance may be associated with the CG configuration. For example, the UE **502** may transmit the CG status indication as part of a CG communication during the CG instance **516**. In some aspects, the last CG instance may be associated with an additional CG configuration. For example, as shown, the UE **502** also may be configured, by the network node **504**, with an additional CG configuration **526** associated with CG instances **528**, **530**, **532**, and **534**. The UE **502** may transmit the CG status indication during the CG instance **530**.

In some aspects, the last uplink communication may include a dynamically granted PUSCH communication. The dynamically granted PUSCH communication may indicate an identifier corresponding to the at least one CG instance. In some aspects, the UE **502** may transmit the CG status indication based on transmitting a BSR associated with the at least one CG instance.

As shown by reference number **536**, the UE **502** may transmit, and the network node **504** may receive, one or more CG communications based at least in part on the status of the at least one CG instance. In some aspects, the CG communication may include a PUSCH payload or a piggyback uplink control information (UCI) transmission. In some aspects, the UE **502** may transmit a CG communication associated with at least one additional CG instance (e.g., the CG instance **516** and/or the CG instance **520**) associated with the CG configuration and may receive an accumulated HARQ-ACK communication associated with the at least one additional CG instance, as shown. In some aspects, the UE **502** may not receive a HARQ-ACK communication associated with the skipped CG instance **518**. In some aspects, the UE **502** may receive a HARQ-ACK communication associated with the skipped CG instance **518** but, although the UE **502** may decode a HARQ-ACK indicator, of the accumulated HARQ-ACK communication, corresponding to the at least one additional CG instance **516** and/or **520**, the UE **502** may not decode the HARQ-ACK indicator associated with the skipped CG instance **518**.

As indicated above, FIG. **5** is provided as an example. Other examples may differ from what is described with respect to FIG. **5**.

FIG. **6** is a diagram illustrating an example process **600** performed, for example, by a UE, in accordance with the present disclosure. Example process **600** is an example where the UE (e.g., UE **502**) performs operations associated with status indications for uplink CG instances.

As shown in FIG. **6**, in some aspects, process **600** may include transmitting a CG status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance (block **610**). For example, the UE (e.g., using communication manager **808** and/or transmission component **804**, depicted in FIG. **8**) may transmit a CG status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance, as described above.

As further shown in FIG. **6**, in some aspects, process **600** may include transmitting one or more CG communications based at least in part on the status of the at least one CG instance (block **620**). For example, the UE (e.g., using communication manager **808** and/or transmission component **804**, depicted in FIG. **8**) may transmit one or more CG communications based at least in part on the status of the at least one CG instance, as described above.

Process **600** may include additional aspects, such as any single aspect or any combination of aspects described below and/or in connection with one or more other processes described elsewhere herein.

In some aspects, process **600** includes determining that a status indication criterion is satisfied, wherein transmitting the CG status indication comprises transmitting the CG status indication based at least in part on determining that the status indication criterion is satisfied. In some aspects, the status indication criterion is satisfied based at least in part on a data size corresponding to an uplink buffer satisfying a data size threshold. In some aspects, the status indication criterion is satisfied based at least in part on an output of a machine learning model associated with incoming data traffic.

In some aspects, the CG status indicates a quantity of CG instances of the at least one CG instance. In some aspects, the CG status indication comprises a skipping indication that indicates at least one skipped CG instance, of a plurality of CG instances that includes the at least one CG instance, during which a CG communication is not to be transmitted. In some aspects, the CG status indicates a quantity of skipped CG instances, of the at least one skipped CG instance. In some aspects, the quantity of skipped CG instances is no greater than a maximum quantity of skipped CG instances. In some aspects, process **600** includes receiving a radio resource control message comprising a configuration that indicates the maximum quantity of skipped CG instances. In some aspects, the maximum quantity of skipped CG instances is based at least in part on a periodicity corresponding to the CG configuration. In some aspects, the UE includes a memory, and the memory includes an indication of a relation between the periodicity and the maximum quantity of skipped CG instances. In some aspects, process **600** includes a radio resource control message comprising a configuration that indicates a mapping between the periodicity and the maximum quantity of skipped CG instances.

In some aspects, the CG status indication corresponds to a specified time period. In some aspects, the specified time period begins at a time instance associated with an offset with respect to transmission of the CG status indication or reception, by a network node, of the CG status indication. In some aspects, transmitting the CG status indication comprises transmitting a CG communication, associated with a last CG instance before the at least one CG instance, wherein the CG communication includes the CG status indication. In some aspects, the CG communication comprises a physical uplink shared channel payload or a piggyback uplink control information transmission. In some aspects, the last CG instance is associated with the CG configuration. In some aspects, the last CG instance is associated with an additional CG configuration. In some aspects, transmitting the CG status indication comprises transmitting a last uplink communication before the at least one CG instance, wherein the last uplink communication includes the CG status indication. In some aspects, the last uplink communication comprises a dynamically granted physical uplink shared channel (PUSCH) communication. In some aspects, the dynamically granted PUSCH communication indicates an identifier corresponding to the at least one CG instance.

In some aspects, process **600** includes transmitting a CG communication associated with at least one additional CG instance associated with the CG configuration, receiving an accumulated HARQ-ACK communication associated with the at least one CG instance and the at least one additional CG instance, decoding a HARQ-ACK indicator, of the accumulated HARQ-ACK communication, corresponding to the at least one additional CG instance, and refraining from decoding a HARQ-ACK indicator, of the accumulated HARQ-ACK communication, corresponding to the at least one CG instance. In some aspects, transmitting the CG status indication comprises transmitting a BSR associated with the at least one CG instance. In some aspects, process **600** includes receiving a BSR configuration that indicates a BSR threshold, wherein the one or more CG communications do not include a CG communication associated with the at least one CG instance based at least in part on a buffer status associated with the at least one CG instance not satisfying the BSR threshold.

Although FIG. **6** shows example blocks of process **600**, in some aspects, process **600** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **6**. Additionally, or alternatively, two or more of the blocks of process **600** may be performed in parallel.

FIG. **7** is a diagram illustrating an example process **700** performed, for example, by a network node, in accordance with the present disclosure. Example process **700** is an example where the network node (e.g., network node **504** performs operations associated with status indications for uplink CG instances.

As shown in FIG. **7**, in some aspects, process **700** may include receiving, from a UE, a CG status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance (block **710**). For example, the network node (e.g., using communication manager **908** and/or reception component **902**, depicted in FIG. **9**) may receive, from a UE, a CG status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance, as described above.

As further shown in FIG. **7**, in some aspects, process **700** may include receiving one or more CG communications based at least in part on the status of the at least one CG instance (block **720**). For example, the network node (e.g., using communication manager **908** and/or reception component **902**, depicted in FIG. **9**) may receive one or more CG communications based at least in part on the status of the at least one CG instance, as described above.

Process **700** may include additional aspects, such as any single aspect or any combination of aspects described below and/or in connection with one or more other processes described elsewhere herein.

In some aspects, receiving the CG status indication comprises receiving the CG status indication based at least in part on a status indication criterion being satisfied. In some aspects, the status indication criterion is satisfied based at least in part on a data size corresponding to an uplink buffer satisfying a data size threshold. In some aspects, the status indication criterion is satisfied based at least in part on an output of a machine learning model associated with incoming data traffic. In some aspects, the CG status indicates a quantity of CG instances of the at least one CG instance. In some aspects, the CG status indication comprises a skipping indication that indicates at least one skipped CG instance, of a plurality of CG instances that includes the at least one CG instance, during which a CG communication is not to be transmitted. In some aspects, the CG status indicates a quantity of skipped CG instances, of the at least one skipped CG instance. In some aspects, the quantity of skipped CG instances is no greater than a maximum quantity of skipped CG instances. In some aspects, process **700** includes transmitting a radio resource control message comprising a configuration that indicates the maximum quantity of skipped CG instances. In some aspects, the maximum quantity of skipped CG instances is based at least in part on a periodicity corresponding to the CG configuration. In some aspects, process **700** includes transmitting a radio resource control message comprising a configuration that indicates a mapping between the periodicity and the maximum quantity of skipped CG instances.

In some aspects, the CG status indication corresponds to a specified time period. In some aspects, the specified time period begins at a time instance associated with an offset with respect to transmission of the CG status indication or reception, by the network node, of the CG status indication. In some aspects, receiving the CG status indication comprises receiving a CG communication, associated with a last CG instance before the at least one CG instance, wherein the CG communication includes the CG status indication. In some aspects, the CG communication comprises a physical uplink shared channel payload or a piggyback uplink control information transmission. In some aspects, the last CG instance is associated with the CG configuration. In some aspects, the last CG instance is associated with an additional CG configuration. In some aspects, receiving the CG status indication comprises receiving a last uplink communication before the at least one CG instance, wherein the last uplink communication includes the CG status indication. In some aspects, the last uplink communication comprises a dynamically granted PUSCH communication. In some aspects, the dynamically granted PUSCH communication indicates an identifier corresponding to the at least one CG instance.

In some aspects, process **700** includes receiving a CG communication associated with at least one additional CG instance associated with the CG configuration, and transmitting an accumulated HARQ-ACK communication associated with the at least one additional CG instance, wherein the accumulated HARQ-ACK communication does not include a HARQ-ACK indicator corresponding to the at least one CG instance. In some aspects, receiving the CG status indication comprises receiving a BSR associated with the at least one CG instance. In some aspects, process **700** includes transmitting a BSR configuration that indicates a BSR threshold, wherein the one or more CG communications do not include a CG communication associated with the at least one CG instance based at least in part on a buffer status associated with the at least one CG instance not satisfying the BSR threshold.

Although FIG. **7** shows example blocks of process **700**, in some aspects, process **700** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **7**. Additionally, or alternatively, two or more of the blocks of process **700** may be performed in parallel.

FIG. **8** is a diagram of an example apparatus **800** for wireless communication, in accordance with the present disclosure. The apparatus **800** may be a UE, or a UE may include the apparatus **800**. In some aspects, the apparatus **800** includes a reception component **802** and a transmission component **804**, which may be in communication with one another (for example, via one or more buses and/or one or more other components). As shown, the apparatus **800** may communicate with another apparatus **806** (such as a UE, a base station, or another wireless communication device) using the reception component **802** and the transmission component **804**. As further shown, the apparatus **800** may include a communication manager **808**. The communication manager **808** may include a determination component **810**.

In some aspects, the apparatus **800** may be configured to perform one or more operations described herein in connection with FIG. **5**. Additionally, or alternatively, the apparatus **800** may be configured to perform one or more processes described herein, such as process **600** of FIG. **6**. In some aspects, the apparatus **800** and/or one or more components shown in FIG. **8** may include one or more components of the UE described in connection with FIG. **2**. Additionally, or alternatively, one or more components shown in FIG. **8** may be implemented within one or more components described in connection with FIG. **2**. Additionally, or alternatively, one or more components of the set of components may be implemented at least in part as software stored in a memory. For example, a component (or a portion of a component) may be implemented as instructions or code stored in a non-transitory computer-readable medium and executable by a controller or a processor to perform the functions or operations of the component.

The reception component **802** may receive communications, such as reference signals, control information, data communications, or a combination thereof, from the apparatus **806**. The reception component **802** may provide received communications to one or more other components of the apparatus **800**. In some aspects, the reception component **802** may perform signal processing on the received communications (such as filtering, amplification, demodulation, analog-to-digital conversion, demultiplexing, deinterleaving, de-mapping, equalization, interference cancellation, or decoding, among other examples), and may provide the processed signals to the one or more other components of the apparatus **800**. In some aspects, the reception component **802** may include one or more antennas, a modem, a demodulator, a MIMO detector, a receive processor, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. **2**.

The transmission component **804** may transmit communications, such as reference signals, control information, data communications, or a combination thereof, to the apparatus **806**. In some aspects, one or more other components of the apparatus **800** may generate communications and may provide the generated communications to the transmission component **804** for transmission to the apparatus **806**. In some aspects, the transmission component **804** may perform signal processing on the generated communications (such as filtering, amplification, modulation, digital-to-analog conversion, multiplexing, interleaving, mapping, or encoding, among other examples), and may transmit the processed signals to the apparatus **806**. In some aspects, the transmission component **804** may include one or more antennas, a modem, a modulator, a transmit MIMO processor, a transmit processor, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. **2**. In some aspects, the transmission component **804** may be co-located with the reception component **802** in a transceiver.

The communication manager **808** and/or the transmission component **804** may transmit a CG status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance. In some aspects, the communication manager **808** may include one or more antennas, a modem, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. **2**. In some aspects, the communication manager **808** may include the reception component **802** and/or the transmission component **804**. In some aspects, the communication manager **808** may be, be similar to, include, or be included in, the communication manager **140** depicted in FIGS. **1** and **2**. The communication manager **808** and/or the transmission component **804** may transmit one or more CG communications based at least in part on the status of the at least one CG instance.

The communication manager **808** and/or the determination component **810** may determine that a status indication criterion is satisfied, wherein transmitting the CG status indication comprises transmitting the CG status indication based at least in part on determining that the status indication criterion is satisfied. In some aspects, the determination component **810** may include one or more antennas, a modem, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. **2**. In some aspects, the determination component **810** may include the reception component **802** and/or the transmission component **804**. The communication manager **808** and/or the reception component **802** may receive a radio resource control message comprising a configuration that indicates the maximum quantity of skipped CG instances. The communication manager **808** and/or the transmission component **804** may transmit a CG communication associated with at least one additional CG instance associated with the CG configuration.

The communication manager **808** and/or the reception component **802** may receive an accumulated HARQ-ACK communication associated with the at least one CG instance and the at least one additional CG instance. The communication manager **808** and/or the reception component **802** may decode a HARQ-ACK indicator, of the accumulated HARQ-ACK communication, corresponding to the at least one additional CG instance. The communication manager **808** and/or the reception component **802** may refrain from decoding a HARQ-ACK indicator, of the accumulated HARQ-ACK communication, corresponding to the at least one CG instance. The communication manager **808** and/or the reception component **802** may receive a BSR configuration that indicates a BSR threshold, wherein the one or more CG communications do not include a CG communication associated with the at least one CG instance based at least in part on a buffer status associated with the at least one CG instance not satisfying the BSR threshold.

The number and arrangement of components shown in FIG. **8** are provided as an example. In practice, there may be additional components, fewer components, different components, or differently arranged components than those shown in FIG. **8**. Furthermore, two or more components shown in FIG. **8** may be implemented within a single component, or a single component shown in FIG. **8** may be implemented as multiple, distributed components. Additionally, or alternatively, a set of (one or more) components shown in FIG. **8** may perform one or more functions described as being performed by another set of components shown in FIG. **8**.

FIG. **9** is a diagram of an example apparatus **900** for wireless communication, in accordance with the present disclosure. The apparatus **900** may be a network node, or a network node may include the apparatus **900**. In some aspects, the apparatus **900** includes a reception component **902** and a transmission component **904**, which may be in communication with one another (for example, via one or more buses and/or one or more other components). As shown, the apparatus **900** may communicate with another apparatus **906** (such as a UE, a base station, or another wireless communication device) using the reception component **902** and the transmission component **904**. As further shown, the apparatus **900** may include a communication manager **908**.

In some aspects, the apparatus **900** may be configured to perform one or more operations described herein in connection with FIG. **5**. Additionally, or alternatively, the apparatus **900** may be configured to perform one or more processes described herein, such as process **700** of FIG. **7**. In some aspects, the apparatus **900** and/or one or more components shown in FIG. **9** may include one or more components of the network node described in connection with FIG. **2**. Additionally, or alternatively, one or more components shown in FIG. **9** may be implemented within one or more components described in connection with FIG. **2**. Additionally, or alternatively, one or more components of the set of components may be implemented at least in part as software stored in a memory. For example, a component (or a portion of a component) may be implemented as instructions or code stored in a non-transitory computer-readable medium and executable by a controller or a processor to perform the functions or operations of the component.

The reception component **902** may receive communications, such as reference signals, control information, data communications, or a combination thereof, from the apparatus **906**. The reception component **902** may provide received communications to one or more other components of the apparatus **900**. In some aspects, the reception component **902** may perform signal processing on the received communications (such as filtering, amplification, demodulation, analog-to-digital conversion, demultiplexing, deinterleaving, de-mapping, equalization, interference cancellation, or decoding, among other examples), and may provide the processed signals to the one or more other components of the apparatus **900**. In some aspects, the reception component **902** may include one or more antennas, a modem, a demodulator, a MIMO detector, a receive processor, a controller/processor, a memory, or a combination thereof, of the network node described in connection with FIG. **2**.

The transmission component **904** may transmit communications, such as reference signals, control information, data communications, or a combination thereof, to the apparatus **906**. In some aspects, one or more other components of the apparatus **900** may generate communications and may provide the generated communications to the transmission component **904** for transmission to the apparatus **906**. In some aspects, the transmission component **904** may perform signal processing on the generated communications (such as filtering, amplification, modulation, digital-to-analog conversion, multiplexing, interleaving, mapping, or encoding, among other examples), and may transmit the processed signals to the apparatus **906**. In some aspects, the transmission component **904** may include one or more antennas, a modem, a modulator, a transmit MIMO processor, a transmit processor, a controller/processor, a memory, or a combination thereof, of the network node described in connection with FIG. **2**. In some aspects, the transmission component **904** may be co-located with the reception component **902** in a transceiver.

The communication manager **908** and/or the reception component **902** may receive, from a UE, a CG status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance. In some aspects, the communication manager **908** may include one or more antennas, a modem, a controller/processor, a memory, or a combination thereof, of the network node described in connection with FIG. **2**. In some aspects, the communication manager **908** may include the reception component **902** and/or the transmission component **904**. In some aspects, the communication manager **908** may be, be similar to, include, or be included in, the communication manager **150** depicted in FIGS. **1** and **2**. The communication manager **908** and/or the reception component **902** may receive one or more CG communications based at least in part on the status of the at least one CG instance.

The communication manager **908** and/or the transmission component **904** may transmit a radio resource control message comprising a configuration that indicates the maximum quantity of skipped CG instances. The communication manager **908** and/or the transmission component **904** may transmit a radio resource control message comprising a configuration that indicates a mapping between the periodicity and the maximum quantity of skipped CG instances. The communication manager **908** and/or the reception component **902** may receive a CG communication associated with at least one additional CG instance associated with the CG configuration.

The communication manager **908** and/or the transmission component **904** may transmit an accumulated HARQ-ACK communication associated with the at least one additional CG instance, wherein the accumulated HARQ-ACK communication does not include a HARQ-ACK indicator corresponding to the at least one CG instance. The communication manager **908** and/or the transmission component **904** may transmit a BSR configuration that indicates a BSR threshold, wherein the one or more CG communications do not include a CG communication associated with the at least one CG instance based at least in part on a buffer status associated with the at least one CG instance not satisfying the BSR threshold.

The number and arrangement of components shown in FIG. **9** are provided as an example. In practice, there may be additional components, fewer components, different components, or differently arranged components than those shown in FIG. **9**. Furthermore, two or more components shown in FIG. **9** may be implemented within a single component, or a single component shown in FIG. **9** may be implemented as multiple, distributed components. Additionally, or alternatively, a set of (one or more) components shown in FIG. **9** may perform one or more functions described as being performed by another set of components shown in FIG. **9**.

The following provides an overview of some Aspects of the present disclosure:

Aspect 1: A method of wireless communication performed by a user equipment (UE), comprising: transmitting a configured grant (CG) status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance; and transmitting one or more CG communications based at least in part on the status of the at least one CG instance.

Aspect 2: The method of Aspect 1, further comprising determining that a status indication criterion is satisfied, wherein transmitting the CG status indication comprises transmitting the CG status indication based at least in part on determining that the status indication criterion is satisfied.

Aspect 3: The method of Aspect 2, wherein the status indication criterion is satisfied based at least in part on a data size corresponding to an uplink buffer satisfying a data size threshold.

Aspect 4: The method of either of Aspects 2 or 3, wherein the status indication criterion is satisfied based at least in part on an output of a machine learning model associated with incoming data traffic.

Aspect 5: The method of any of Aspects 1-4, wherein the CG status indicates a quantity of CG instances of the at least one CG instance.

Aspect 6: The method of any of Aspects 1-5, wherein the CG status indication comprises a skipping indication that indicates at least one skipped CG instance, of a plurality of CG instances that includes the at least one CG instance, during which a CG communication is not to be transmitted.

Aspect 7: The method of Aspect 6, wherein the CG status indicates a quantity of skipped CG instances, of the at least one skipped CG instance.

Aspect 8: The method of Aspect 7, wherein the quantity of skipped CG instances is no greater than a maximum quantity of skipped CG instances.

Aspect 9: The method of Aspect 8, further comprising receiving a radio resource control message comprising a configuration that indicates the maximum quantity of skipped CG instances.

Aspect 10: The method of either of Aspects 8 or 9, wherein the maximum quantity of skipped CG instances is based at least in part on a periodicity corresponding to the CG configuration.

Aspect 11: The method of Aspect 10, wherein the UE includes a memory, and wherein the memory includes an indication of a relation between the periodicity and the maximum quantity of skipped CG instances.

Aspect 12: The method of either of Aspects 10 or 11, further comprising a radio resource control message comprising a configuration that indicates a mapping between the periodicity and the maximum quantity of skipped CG instances.

Aspect 13: The method of any of Aspects 1-12, wherein the CG status indication corresponds to a specified time period.

Aspect 14: The method of Aspect 13, wherein the specified time period begins at a time instance associated with an offset with respect to transmission of the CG status indication or reception, by a network node, of the CG status indication.

Aspect 15: The method of any of Aspects 1-14, wherein transmitting the CG status indication comprises transmitting a CG communication, associated with a last CG instance before the at least one CG instance, wherein the CG communication includes the CG status indication.

Aspect 16: The method of Aspect 15, wherein the CG communication comprises a physical uplink shared channel payload or a piggyback uplink control information transmission.

Aspect 17: The method of either of Aspects 15 or 16, wherein the last CG instance is associated with the CG configuration.

Aspect 18: The method of any of Aspects 15-17, wherein the last CG instance is associated with an additional CG configuration.

Aspect 19: The method of any of Aspects 1-18, wherein transmitting the CG status indication comprises transmitting a last uplink communication before the at least one CG instance, wherein the last uplink communication includes the CG status indication.

Aspect 20: The method of Aspect 19, wherein the last uplink communication comprises a dynamically granted physical uplink shared channel (PUSCH) communication.

Aspect 21: The method of Aspect 20, wherein the dynamically granted PUSCH communication indicates an identifier corresponding to the at least one CG instance.

Aspect 22: The method of any of Aspects 1-21, further comprising: transmitting a CG communication associated with at least one additional CG instance associated with the CG configuration; receiving an accumulated hybrid automatic response request (HARQ)-acknowledgment (HARQ-ACK) communication associated with the at least one CG instance and the at least one additional CG instance; decoding a HARQ-ACK indicator, of the accumulated HARQ-ACK communication, corresponding to the at least one additional CG instance; and refraining from decoding a HARQ-ACK indicator, of the accumulated HARQ-ACK communication, corresponding to the at least one CG instance.

Aspect 23: The method of any of Aspects 1-22, wherein transmitting the CG status indication comprises transmitting a buffer status report (BSR) associated with the at least one CG instance.

Aspect 24: The method of Aspect 23, further comprising receiving a BSR configuration that indicates a BSR threshold, wherein the one or more CG communications do not include a CG communication associated with the at least one CG instance based at least in part on a buffer status associated with the at least one CG instance not satisfying the BSR threshold.

Aspect 25: A method of wireless communication performed by a network node, comprising: receiving, from a user equipment (UE), a configured grant (CG) status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance; and receiving one or more CG communications based at least in part on the status of the at least one CG instance.

Aspect 26: The method of Aspect 25, wherein receiving the CG status indication comprises receiving the CG status indication based at least in part on a status indication criterion being satisfied.

Aspect 27: The method of Aspect 26, wherein the status indication criterion is satisfied based at least in part on a data size corresponding to an uplink buffer satisfying a data size threshold.

Aspect 28: The method of either of Aspects 26 or 27, wherein the status indication criterion is satisfied based at least in part on an output of a machine learning model associated with incoming data traffic.

Aspect 29: The method of any of Aspects 25-28, wherein the CG status indicates a quantity of CG instances of the at least one CG instance.

Aspect 30: The method of any of Aspects 25-29, wherein the CG status indication comprises a skipping indication that indicates at least one skipped CG instance, of a plurality of CG instances that includes the at least one CG instance, during which a CG communication is not to be transmitted.

Aspect 31: The method of Aspect 30, wherein the CG status indicates a quantity of skipped CG instances, of the at least one skipped CG instance.

Aspect 32: The method of Aspect 31, wherein the quantity of skipped CG instances is no greater than a maximum quantity of skipped CG instances.

Aspect 33: The method of Aspect 32, further comprising transmitting a radio resource control message comprising a configuration that indicates the maximum quantity of skipped CG instances.

Aspect 34: The method of either of Aspects 32 or 33, wherein the maximum quantity of skipped CG instances is based at least in part on a periodicity corresponding to the CG configuration.

Aspect 35: The method of Aspect 34, further comprising transmitting a radio resource control message comprising a configuration that indicates a mapping between the periodicity and the maximum quantity of skipped CG instances.

Aspect 36: The method of any of Aspects 25-35, wherein the CG status indication corresponds to a specified time period.

Aspect 37: The method of Aspect 36, wherein the specified time period begins at a time instance associated with an offset with respect to transmission of the CG status indication or reception, by the network node, of the CG status indication.

Aspect 38: The method of any of Aspects 25-37, wherein receiving the CG status indication comprises receiving a CG communication, associated with a last CG instance before the at least one CG instance, wherein the CG communication includes the CG status indication.

Aspect 39: The method of Aspect 38, wherein the CG communication comprises a physical uplink shared channel payload or a piggyback uplink control information transmission.

Aspect 40: The method of either of Aspects 38 or 39, wherein the last CG instance is associated with the CG configuration.

Aspect 41: The method of any of Aspects 38-40, wherein the last CG instance is associated with an additional CG configuration.

Aspect 42: The method of any of Aspects 25-41, wherein receiving the CG status indication comprises receiving a last uplink communication before the at least one CG instance, wherein the last uplink communication includes the CG status indication.

Aspect 43: The method of Aspect 42, wherein the last uplink communication comprises a dynamically granted physical uplink shared channel (PUSCH) communication.

Aspect 44: The method of Aspect 43, wherein the dynamically granted PUSCH communication indicates an identifier corresponding to the at least one CG instance.

Aspect 45: The method of any of Aspects 25-44, further comprising: receiving a CG communication associated with at least one additional CG instance associated with the CG configuration; and transmitting an accumulated hybrid automatic response request (HARQ)-acknowledgment (HARQ-ACK) communication associated with the at least one additional CG instance, wherein the accumulated HARQ-ACK communication does not include a HARQ-ACK indicator corresponding to the at least one CG instance.

Aspect 46: The method of any of Aspects 25-45, wherein receiving the CG status indication comprises receiving a buffer status report (BSR) associated with the at least one CG instance.

Aspect 47: The method of Aspect 46, further comprising transmitting a BSR configuration that indicates a BSR threshold, wherein the one or more CG communications do not include a CG communication associated with the at least one CG instance based at least in part on a buffer status associated with the at least one CG instance not satisfying the BSR threshold.

Aspect 48: An apparatus for wireless communication at a device, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform the method of one or more of Aspects 1-24.

Aspect 49: A device for wireless communication, comprising a memory and one or more processors coupled to the memory, the one or more processors configured to perform the method of one or more of Aspects 1-24.

Aspect 50: An apparatus for wireless communication, comprising at least one means for performing the method of one or more of Aspects 1-24.

Aspect 51: A non-transitory computer-readable medium storing code for wireless communication, the code comprising instructions executable by a processor to perform the method of one or more of Aspects 1-24.

Aspect 52: A non-transitory computer-readable medium storing a set of instructions for wireless communication, the set of instructions comprising one or more instructions that, when executed by one or more processors of a device, cause the device to perform the method of one or more of Aspects 1-24.

Aspect 53: An apparatus for wireless communication at a device, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform the method of one or more of Aspects 25-47.

Aspect 54: A device for wireless communication, comprising a memory and one or more processors coupled to the memory, the one or more processors configured to perform the method of one or more of Aspects 25-47.

Aspect 55: An apparatus for wireless communication, comprising at least one means for performing the method of one or more of Aspects 25-47.

Aspect 56: A non-transitory computer-readable medium storing code for wireless communication, the code comprising instructions executable by a processor to perform the method of one or more of Aspects 25-47.

Aspect 57: A non-transitory computer-readable medium storing a set of instructions for wireless communication, the set of instructions comprising one or more instructions that, when executed by one or more processors of a device, cause the device to perform the method of one or more of Aspects 25-47.

The foregoing disclosure provides illustration and description but is not intended to be exhaustive or to limit the aspects to the precise forms disclosed. Modifications and variations may be made in light of the above disclosure or may be acquired from practice of the aspects.

As used herein, the term "component" is intended to be broadly construed as hardware and/or a combination of hardware and software. "Software" shall be construed broadly to mean instructions, instruction sets, code, code segments, program code, programs, subprograms, software modules, applications, software applications, software packages, routines, subroutines, objects, executables, threads of execution, procedures, and/or functions, among other examples, whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise. As used herein, a "processor" is implemented in hardware and/or a combination of hardware and software. It will be apparent that systems and/or methods described herein may be implemented in different forms of hardware and/or a combination of hardware and software. The actual specialized control hardware or software code used to implement these systems and/or methods is not limiting of the aspects. Thus, the operation and behavior of the systems and/or methods are described herein without reference to specific software code, since those skilled in the art will understand that software and hardware can be designed to implement the systems and/or methods based, at least in part, on the description herein.

As used herein, "satisfying a threshold" may, depending on the context, refer to a value being greater than the threshold, greater than or equal to the threshold, less than the threshold, less than or equal to the threshold, equal to the threshold, not equal to the threshold, or the like.

Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of various aspects. Many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. The disclosure of various aspects includes each dependent claim in combination with every other claim in the claim set. As used herein, a phrase referring to "at least one of" a list of items refers to any combination of those items, including single members. As an example, "at least one of: a, b, or c" is intended to cover a, b, c, a+b, a+c, b+c, and a+b+c, as well as any combination with multiples of the same element (e.g., a+a, a+a+a, a+a+b, a+a+c, a+b+b, a+c+c, b+b, b+b+b, b+b+c, c+c, and c+c+c, or any other ordering of a, b, and c).

No element, act, or instruction used herein should be construed as critical or essential unless explicitly described as such. Also, as used herein, the articles "a" and "an" are intended to include one or more items and may be used interchangeably with "one or more." Further, as used herein, the article "the" is intended to include one or more items referenced in connection with the article "the" and may be used interchangeably with "the one or more." Furthermore, as used herein, the terms "set" and "group" are intended to include one or more items and may be used interchangeably with "one or more." Where only one item is intended, the phrase "only one" or similar language is used. Also, as used herein, the terms "has," "have," "having," or the like are intended to be open-ended terms that do not limit an element that they modify (e.g., an element "having" A may also have B). Further, the phrase "based on" is intended to mean "based, at least in part, on" unless explicitly stated otherwise. Also, as used herein, the term "or" is intended to be inclusive when used in a series and may be used interchangeably with "and/or," unless explicitly stated otherwise (e.g., if used in combination with "either" or "only one of").

What is claimed is:

1. A user equipment (UE) for wireless communication, comprising:

a memory; and

one or more processors, coupled to the memory, configured to:

transmit a configured grant (CG) status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance, and wherein the CG status indication comprises a skipping indication that indicates at least one skipped CG instance, of a plurality of CG instances that includes the at least one CG instance, during which a CG communication is not to be transmitted; and

transmit one or more CG communications based at least in part on the status of the at least one CG instance.

2. The UE of claim **1**, wherein the one or more processors are further configured to determine that a status indication criterion is satisfied, wherein the one or more processors, to transmit the CG status indication, are configured to transmit the CG status indication based at least in part on determining that the status indication criterion is satisfied.

3. The UE of claim **2**, wherein the status indication criterion is satisfied based at least in part on a data size corresponding to an uplink buffer satisfying a data size threshold.

4. The UE of claim **2**, wherein the status indication criterion is satisfied based at least in part on an output of a machine learning model associated with incoming data traffic.

5. The UE of claim **1**, wherein the CG status indicates a quantity of CG instances of the at least one CG instance.

6. The UE of claim **1**, wherein the CG status indicates a quantity of skipped CG instances, of the at least one skipped CG instance.

7. The UE of claim **6**, wherein the quantity of skipped CG instances is no greater than a maximum quantity of skipped CG instances.

8. The UE of claim **7**, wherein the one or more processors are further configured to receive a radio resource control message comprising a configuration that indicates the maximum quantity of skipped CG instances.

9. The UE of claim **7**, wherein the maximum quantity of skipped CG instances is based at least in part on a periodicity corresponding to the CG configuration.

10. The UE of claim **9**, wherein the UE includes a memory, and wherein the memory includes an indication of a relation between the periodicity and the maximum quantity of skipped CG instances.

11. The UE of claim **9**, wherein the one or more processors are further configured to a radio resource control message comprising a configuration that indicates a mapping between the periodicity and the maximum quantity of skipped CG instances.

12. The UE of claim **1**, wherein the CG status indication corresponds to a specified time period.

13. The UE of claim **12**, wherein the specified time period begins at a time instance associated with an offset with respect to transmission of the CG status indication or reception, by a network node, of the CG status indication.

14. The UE of claim **1**, wherein transmitting the CG status indication comprises transmitting a CG communication, associated with a last CG instance before the at least one CG instance, wherein the CG communication includes the CG status indication.

15. The UE of claim **14**, wherein the CG communication comprises a physical uplink shared channel payload or a piggyback uplink control information transmission.

16. The UE of claim **14**, wherein the last CG instance is associated with the CG configuration.

17. The UE of claim **14**, wherein the last CG instance is associated with an additional CG configuration.

18. The UE of claim **1**, wherein the one or more processors, to transmit the CG status indication, are configured to transmit a last uplink communication before the at least one CG instance, wherein the last uplink communication includes the CG status indication.

19. The UE of claim **18**, wherein the last uplink communication comprises a dynamically granted physical uplink shared channel (PUSCH) communication.

20. The UE of claim **19**, wherein the dynamically granted PUSCH communication indicates an identifier corresponding to the at least one CG instance.

21. The UE of claim **20**, wherein the one or more processors are further configured to:

transmit a CG communication associated with at least one additional CG instance associated with the CG configuration;

receive an accumulated hybrid automatic response request (HARQ)-acknowledgment (HARQ-ACK) communication associated with the at least one CG instance and the at least one additional CG instance;

decode a HARQ-ACK indicator, of the accumulated HARQ-ACK communication, corresponding to the at least one additional CG instance; and

refrain from decoding a HARQ-ACK indicator, of the accumulated HARQ-ACK communication, corresponding to the at least one CG instance.

22. The UE of claim **20**, wherein the one or more processors, to transmit the CG status indication, are configured to transmit a buffer status report (BSR) associated with the at least one CG instance.

23. The UE of claim **22**, wherein the one or more processors are further configured to receive a BSR configuration that indicates a BSR threshold, wherein the one or more CG communications do not include a CG communication associated with the at least one CG instance based at least in part on a buffer status associated with the at least one CG instance not satisfying the BSR threshold.

24. A network node for wireless communication, comprising:

a memory; and

one or more processors, coupled to the memory, configured to:

receive, from a user equipment (UE), a configured grant (CG) status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance, and wherein the CG status indication comprises a skipping indication that indicates at least one skipped CG instance, of a plurality of CG instances that includes the at least one CG instance, during which a CG communication is not to be transmitted; and

receive one or more CG communications based at least in part on the status of the at least one CG instance.

25. A method of wireless communication performed by a user equipment (UE), comprising:

transmitting a configured grant (CG) status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance, and wherein the CG status indication comprises a skipping indication that indicates at least one skipped CG instance, of a plurality of CG instances that includes the at least one CG instance, during which a CG communication is not to be transmitted; and

transmitting one or more CG communications based at least in part on the status of the at least one CG instance.

26. The method of claim **25**, further comprising:

determining that a status indication criterion is satisfied, wherein transmitting the CG status indication comprises transmitting the CG status indication based at least in part on determining that the status indication criterion is satisfied.

27. The method of claim **26**, wherein the status indication criterion is satisfied based at least in part on a data size corresponding to an uplink buffer satisfying a data size threshold.

28. The method of claim **26**, wherein the status indication criterion is satisfied based at least in part on an output of a machine learning model associated with incoming data traffic.

29. The method of claim **25**, wherein the CG status indicates a quantity of CG instances of the at least one CG instance.

30. A method of wireless communication performed by a network node, comprising:

receiving, from a user equipment (UE), a configured grant (CG) status indication that indicates a status of at least one CG instance associated with an uplink CG configuration, wherein the status indicates whether a CG communication is to be transmitted during the at least one CG instance, and wherein the CG status indication comprises a skipping indication that indicates at least one skipped CG instance, of a plurality of CG instances that includes the at least one CG instance, during which a CG communication is not to be transmitted; and

receiving one or more CG communications based at least in part on the status of the at least one CG instance.

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